



Tips & Tricks for Building a Custom Step with SAS® Viya

Hands-on Workshop

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Data Engineering and Decisioning
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#SASInnovate

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2026

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What is a Custom Step?

The screenshot displays the SAS Studio interface. On the left, the 'Steps' pane is open, showing a list of 'Custom Steps' under the 'Name' column. A red box highlights the 'Custom Steps' tab, and another red box highlights the 'GeoDistance with Rounding' step in the list. A callout box points to this list with the text: 'Listing of available Custom Steps'. In the center, a flow diagram shows an 'Import File' step connected to a 'GeoDistance with Rounding' step, both highlighted with red boxes. A callout box points to the 'GeoDistance with Rounding' step in the flow with the text: 'Custom Steps can be implemented in a SAS Studio Flow or used Stand-alone.' The main editor area shows the configuration for the 'GeoDistance with Rounding' step, including options for 'Calculate distance in:' (Miles selected) and 'Lat/Long Columns' (country_latitude and country_longitude selected).

Listing of available Custom Steps

Custom Steps can be implemented in a **SAS Studio Flow** or used **Stand-alone**.

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Custom Steps enable you to create a user interface to complete a specific task. Available custom steps are listed on the **Steps** pane in **SAS Studio** on the **Custom Steps** tab.

They can then be used in multiple SAS Studio Flows or stand-alone to ensure a common, repeatable process.

Designing a Custom Step

The screenshot displays the SAS Studio Analyst interface for designing a custom step. The 'Design' tab is selected, showing a 'Control Library' on the left, a 'Page Design Canvas' in the center, and 'Properties' on the right. Annotations highlight key features:

- Control library for adding pages and various controls**: Points to the 'Control Library' on the left.
- Pages**: Points to the 'Data', 'Options', and 'About' tabs at the top of the canvas.
- Page Design Canvas**: Points to the central area where controls are added.
- SAS Studio Analyst license required to use *Design* tab**: A note at the bottom center.
- Control ID must be unique**: Points to the 'ID' field in the 'Properties' panel.
- Properties for the selected control**: Points to the 'Properties' panel on the right.

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The **Design** tab is a visual interface to help you quickly build the JSON for the user interface for your custom step. This JSON is visible on the **JSON** tab. A **SAS Studio Analyst** license is required to use the **Design** tab; if you just have a **SAS Studio Basic** license, then you will need to design your custom steps using only the **JSON** tab.

On the left-hand side is the Control library for adding **pages** and **controls** to the user interface. **Pages** become tabs in the user interface for the custom step.

Each page has its own page design that you customize by adding controls to the **Page Design Canvas**.

For each page or control their **properties** are set on the right-hand side. Each control must have a unique ID within the Custom Step.

Building Custom Steps Tips & Tricks



- 1 Review the custom step **sample controls**
- 2 Review the custom step **starter templates**
- 3 Search the **Custom Step GitHub Repository**
- 4 Test the custom step using **stand-alone** mode
- 5 Review the **log** for macro variable names and values
- 6 Use **%put** statements to aid in the testing of your code
- 7 Use **%global** statement to assign macro variable values for use outside of the custom step
- 8 Use the **New column** control (if your code creates a new column)
- 9 Use **Port Details** to control the output columns
- 10 Create an **About** tab for the custom step

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1. The **sample controls** provide an explanation for each of the control types you can use when designing a custom step.
2. The **starter templates** are example custom steps with associated program code, so you can review some real-life examples of how they were built.
3. Search the **Custom Step GitHub Repository** for any similar custom steps that you may be trying to build and download them to use as a starting point for your design. <https://github.com/sassoftware/sas-studio-custom-steps>
4. Use the **stand-alone** mode to quickly test your custom step.
5. Review the **log** for the macro variable names and their respective values.
6. Use **%put** statements to aid in the testing of your code. This allows you to write out values or show that you have reached a particular portion of the code.
7. Use a **%global** statement to assign macro variable values for use outside of the custom step, if needed.
8. Use the **New column** control if the code within your custom step creates a new column. You can hide this column if you don't want the

end user to change the name or you can choose to display it if you want to give the end user the opportunity to name the new column.

9. Use **Port Details** to control the output columns from the custom step. If removing columns, you also need to remove them in the code of the custom step.
10. Create an **About** tab to provide a description of the custom step and its release date information.

Tips & Tricks for Building Custom Steps with SAS® Viya

Hands-on Exercise

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[SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya: Tips & Tricks for Building a Custom Step in SAS Viya](https://github.com/SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya)

<https://github.com/SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya>

Tips & Tricks for Building a Custom Step with SAS Viya

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Exercise Description

In this hands-on exercise, you will review the top 10 tips and tricks for creating a custom step in SAS Studio.

Log in to SAS Viya

Open a new window in the *Google Chrome* browser and select the **SASLanding** bookmark.

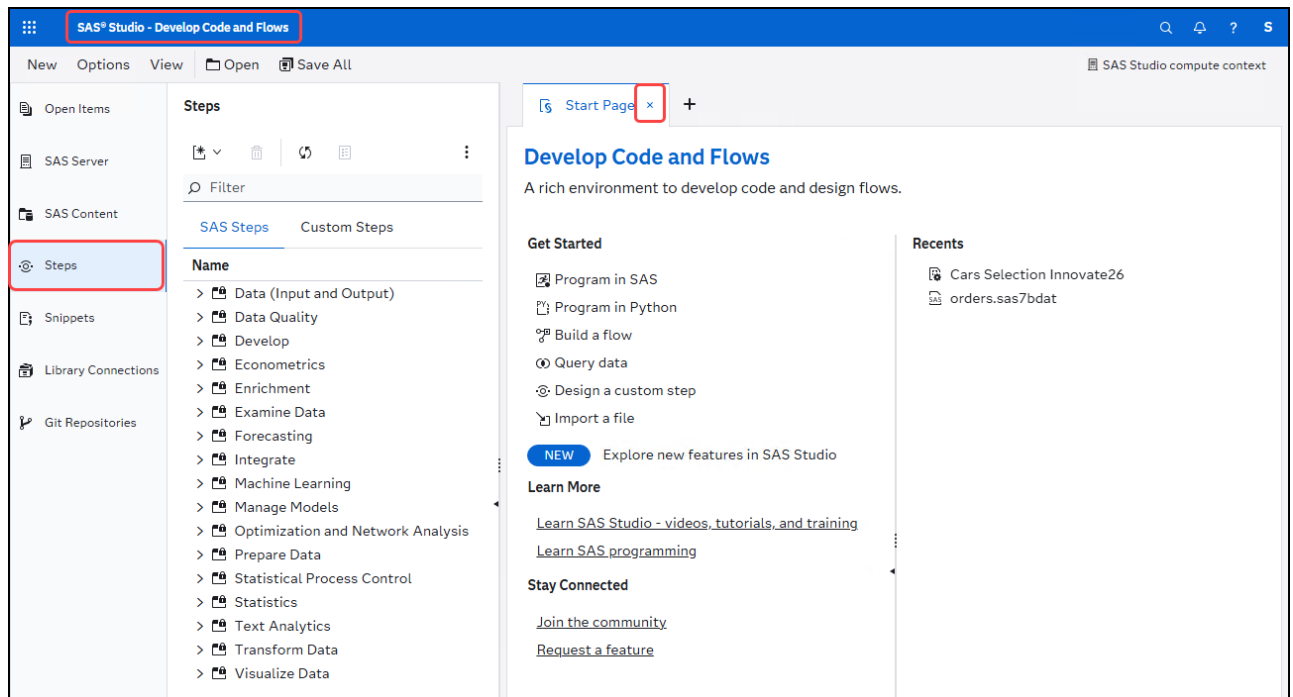
- ID: **student**
- Password: **Metadata0**

Select **No** when prompted about accepting *Admin* privileges.

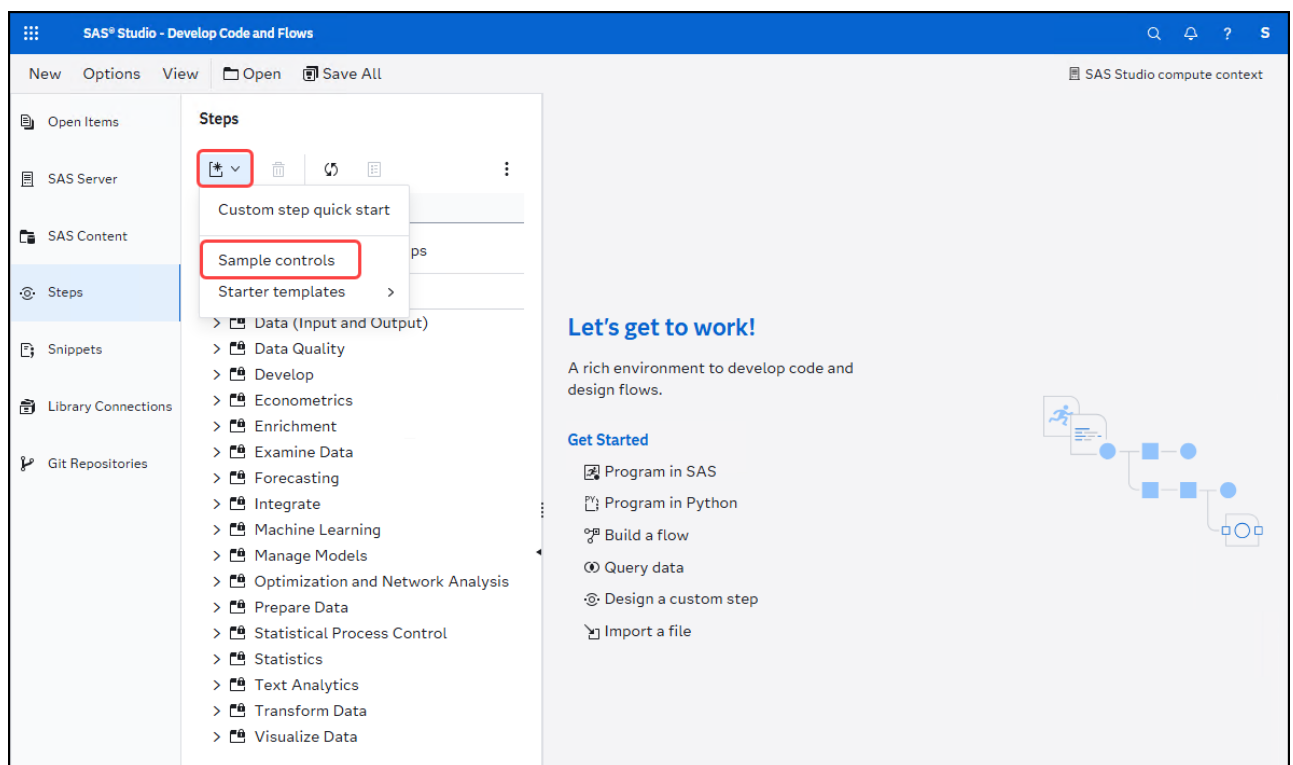
Top 10 Tips & Tricks for Building a Custom Step

1. Review the Custom Step Sample Controls

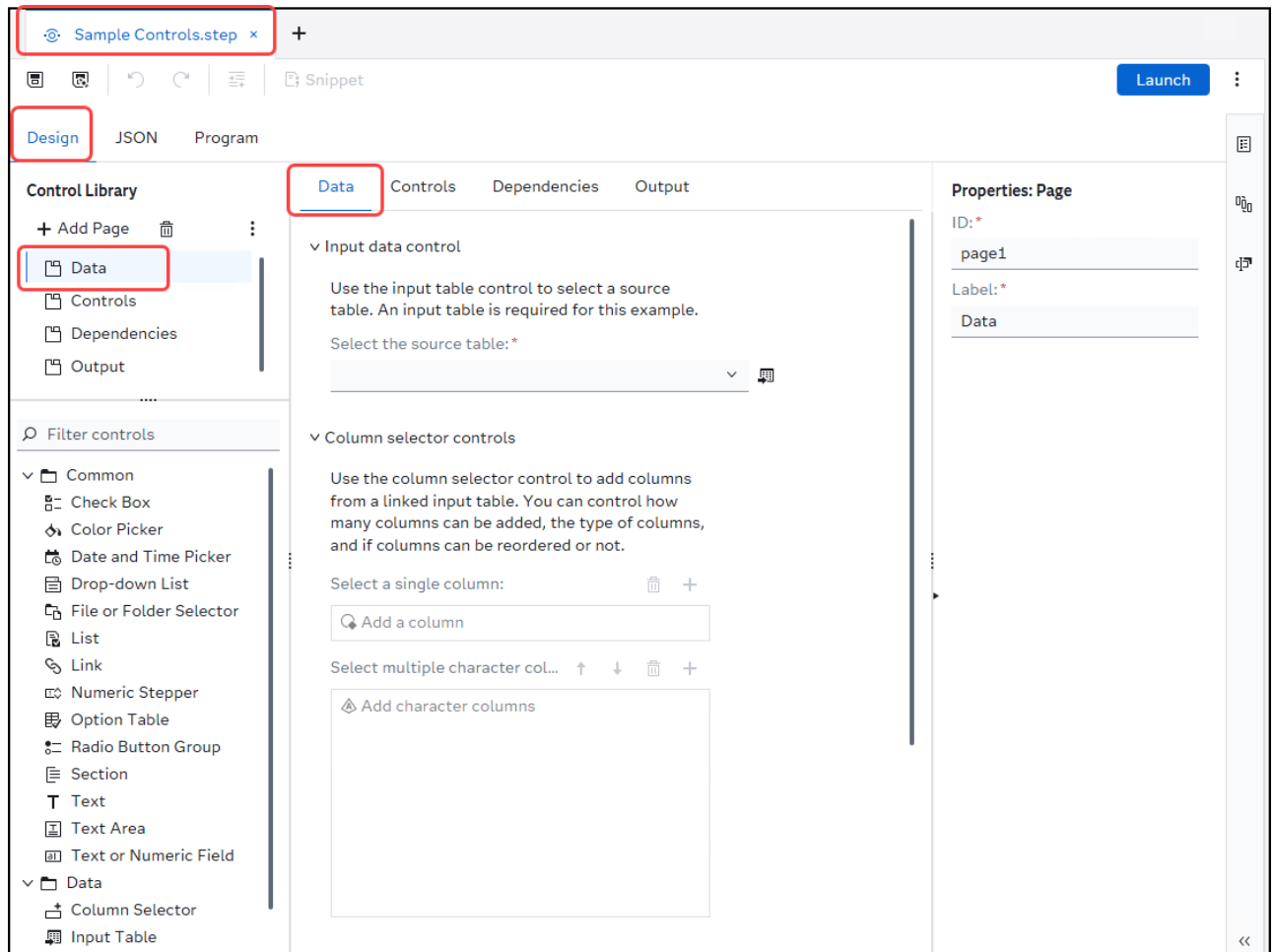
1. Select  → **Develop Code and Flows** to open *SAS Studio*.
2. Click **x** to close the *Start Page*, if it is open. Also, close any other open tabs, if needed.
3. Select  to view the **Steps** pane.



4. On the *Steps* pane, select  → **Sample controls**.

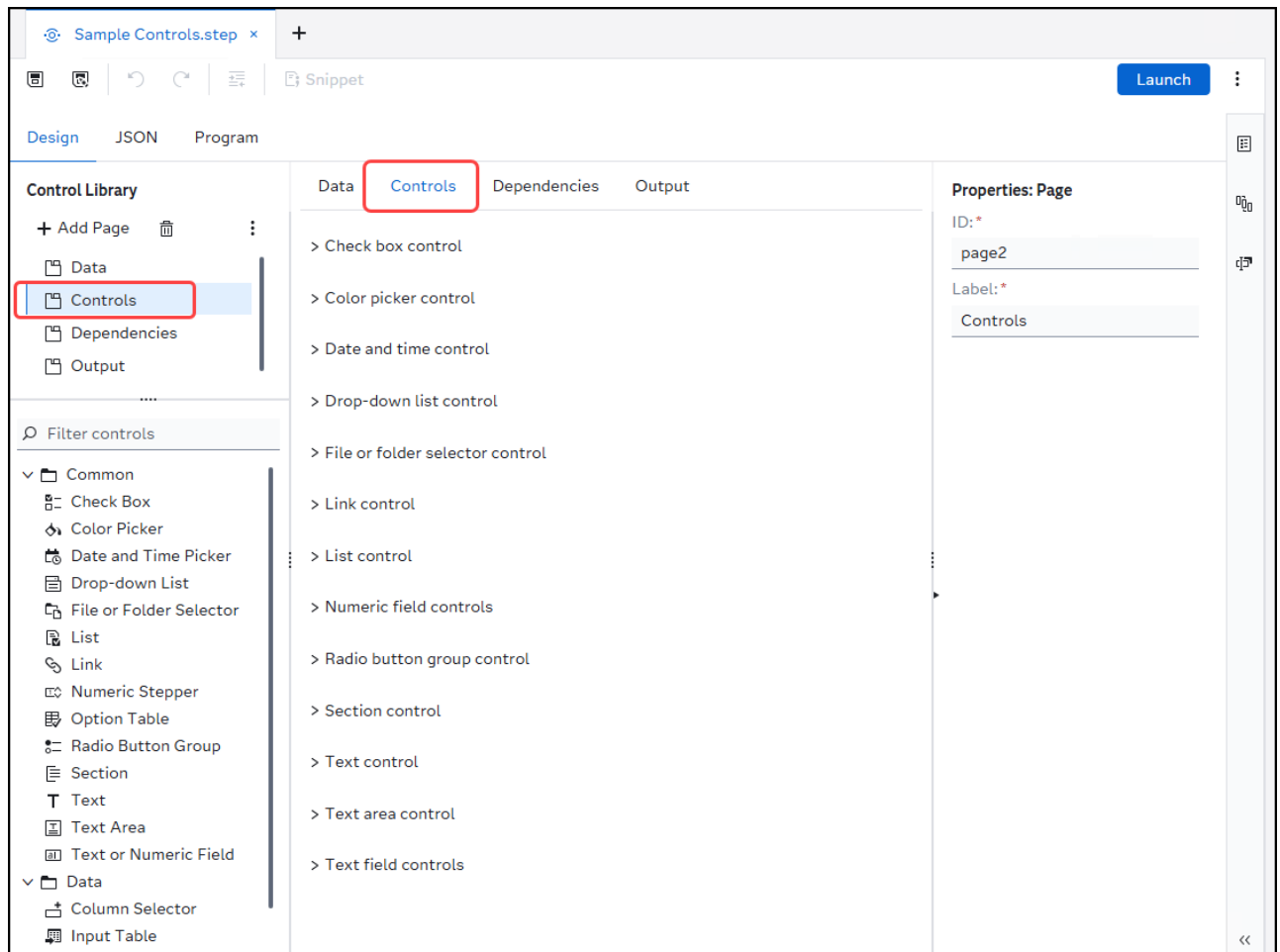


5. On the *Design* tab, select **Data** in the *Control Library* section to view the *Data* page of the *Sample Controls* custom step.



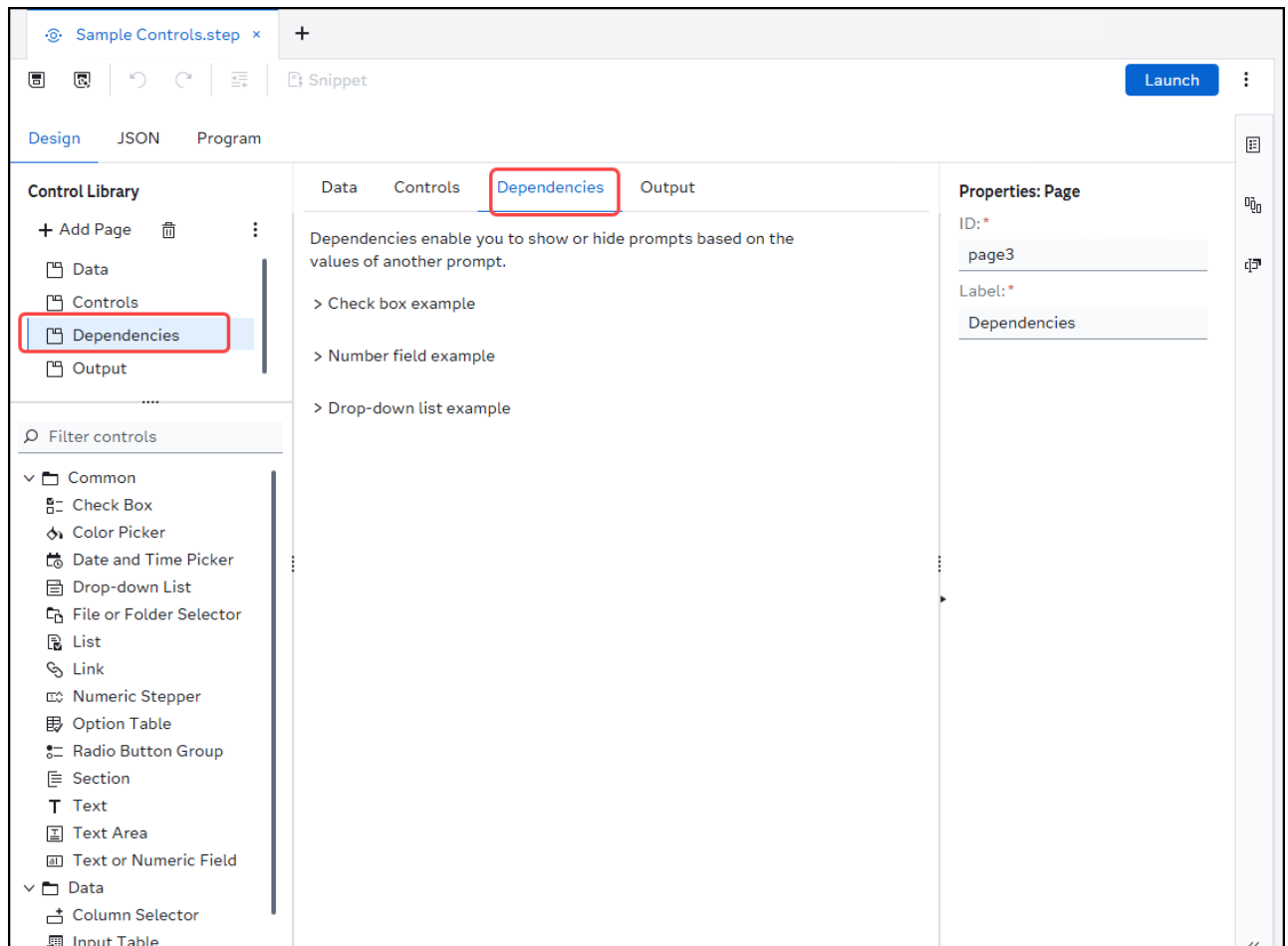
There is an example and text explanation for each of the **Data** controls with the exception of the *Output Table* data control. There is an example and text explanation of that control on a another page.

6. Select **Controls** in the *Control Library* section to view the *Controls* page of the *Sample Controls* custom step.



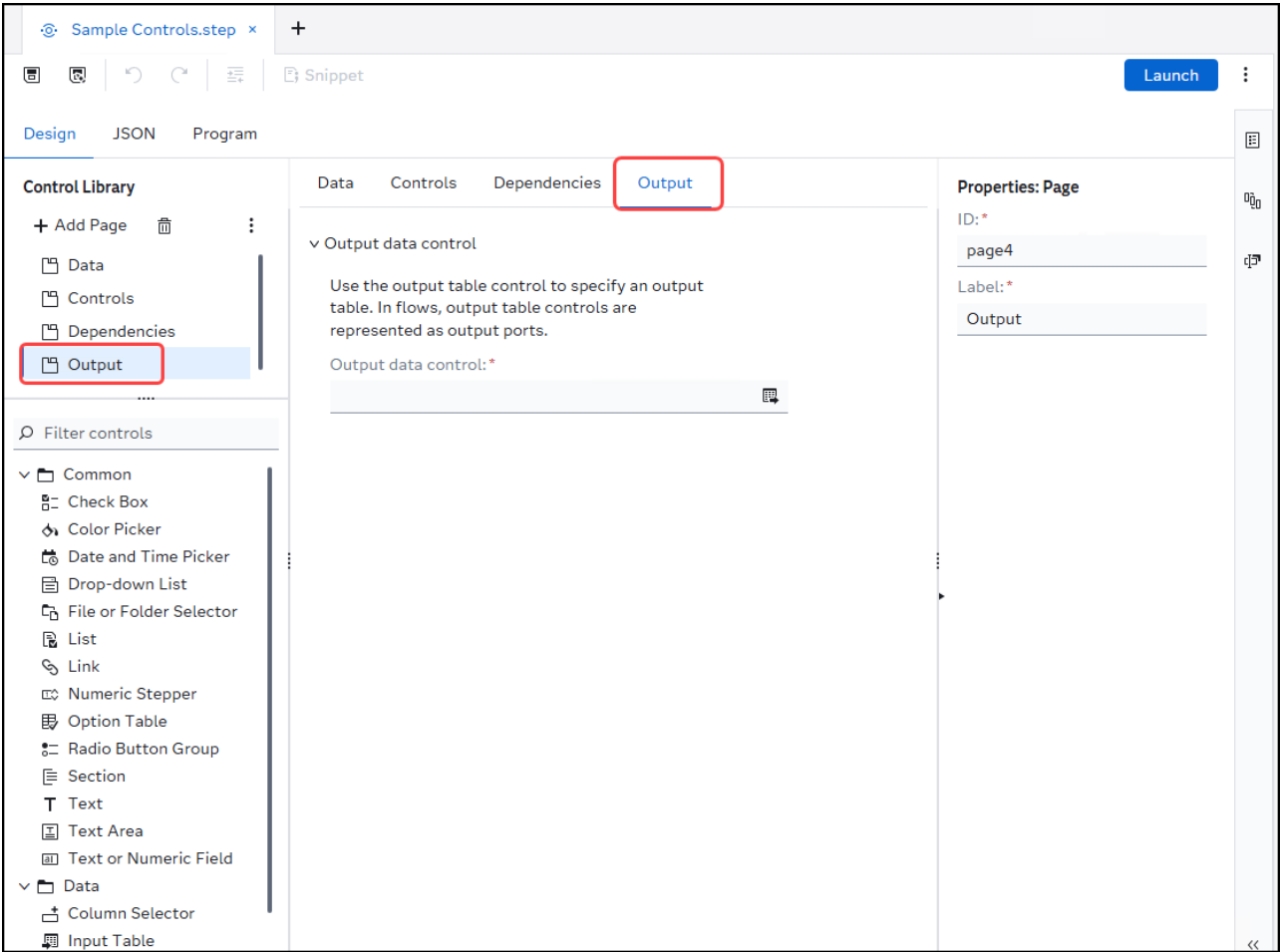
There is an example and text explanation for each of the **Common** controls. Select > to expand the control to view each example and text explanation.

7. Select **Dependencies** in the *Control Library* section to view the *Dependencies* page of the *Sample Controls* custom step.



✎ There is an example and text explanation for dependencies with *check box*, *number field*, and *drop-down list*. Select > to expand the control to view each example and text explanation.

8. Select **Output** in the *Control Library* section to view the *Output* page of the *Sample Controls* custom step.

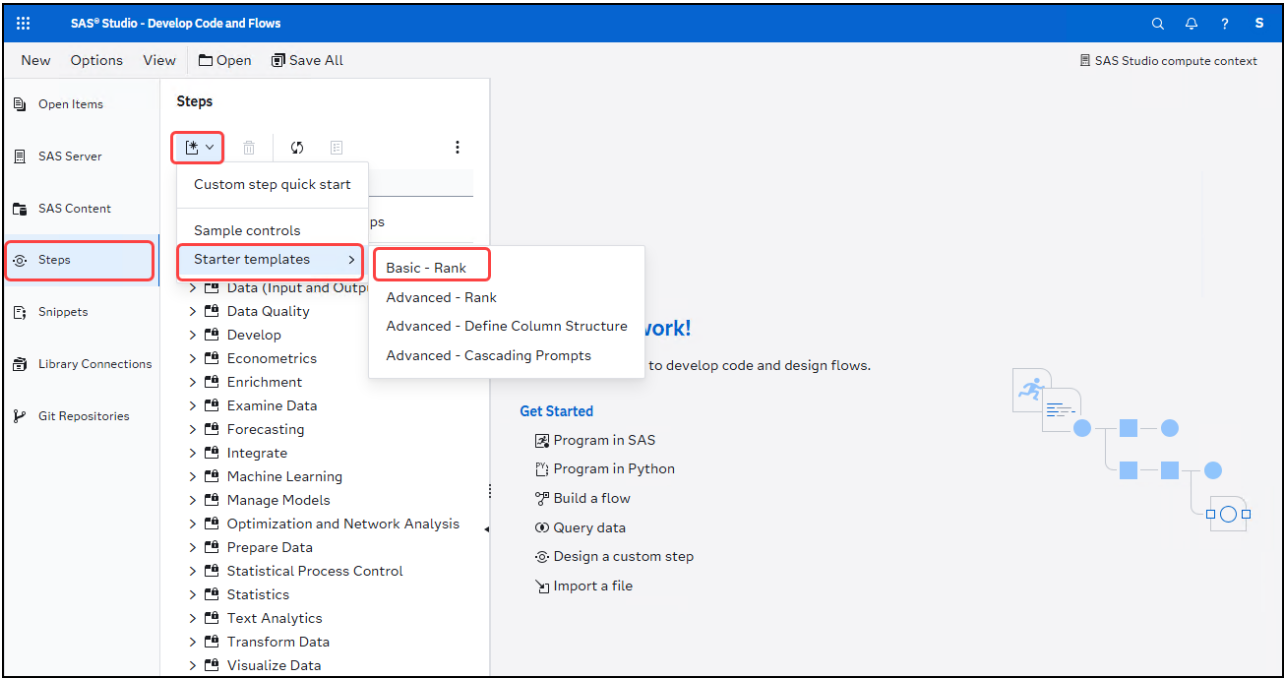


✎ There is an example and text explanation for **Output table** *Data* control.

9. Click **x** to close the *Sample Controls.step* tab.

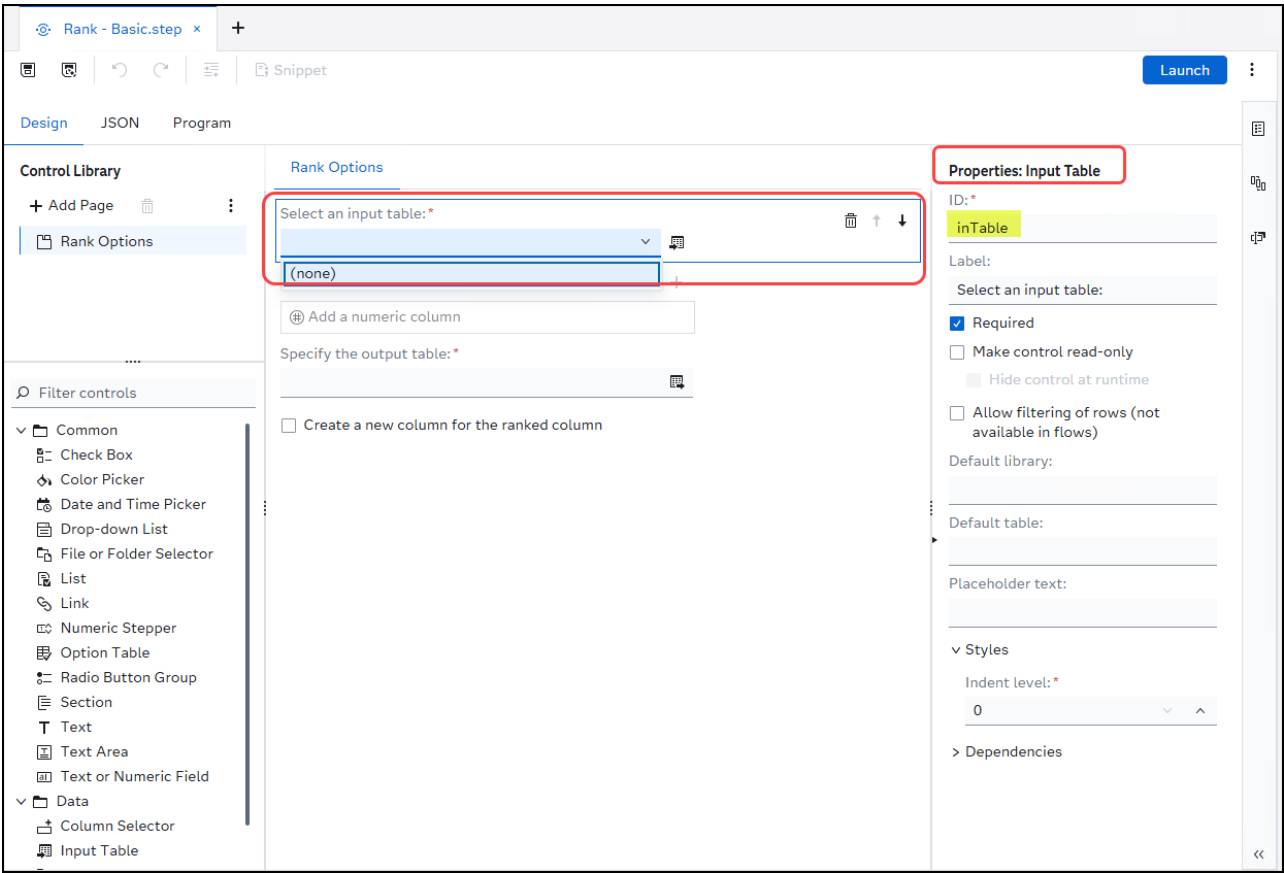
2. Review the Custom Step Starter Templates

- 1. On the *Steps* pane, select  → **Starter templates** → **Basic - Rank**.



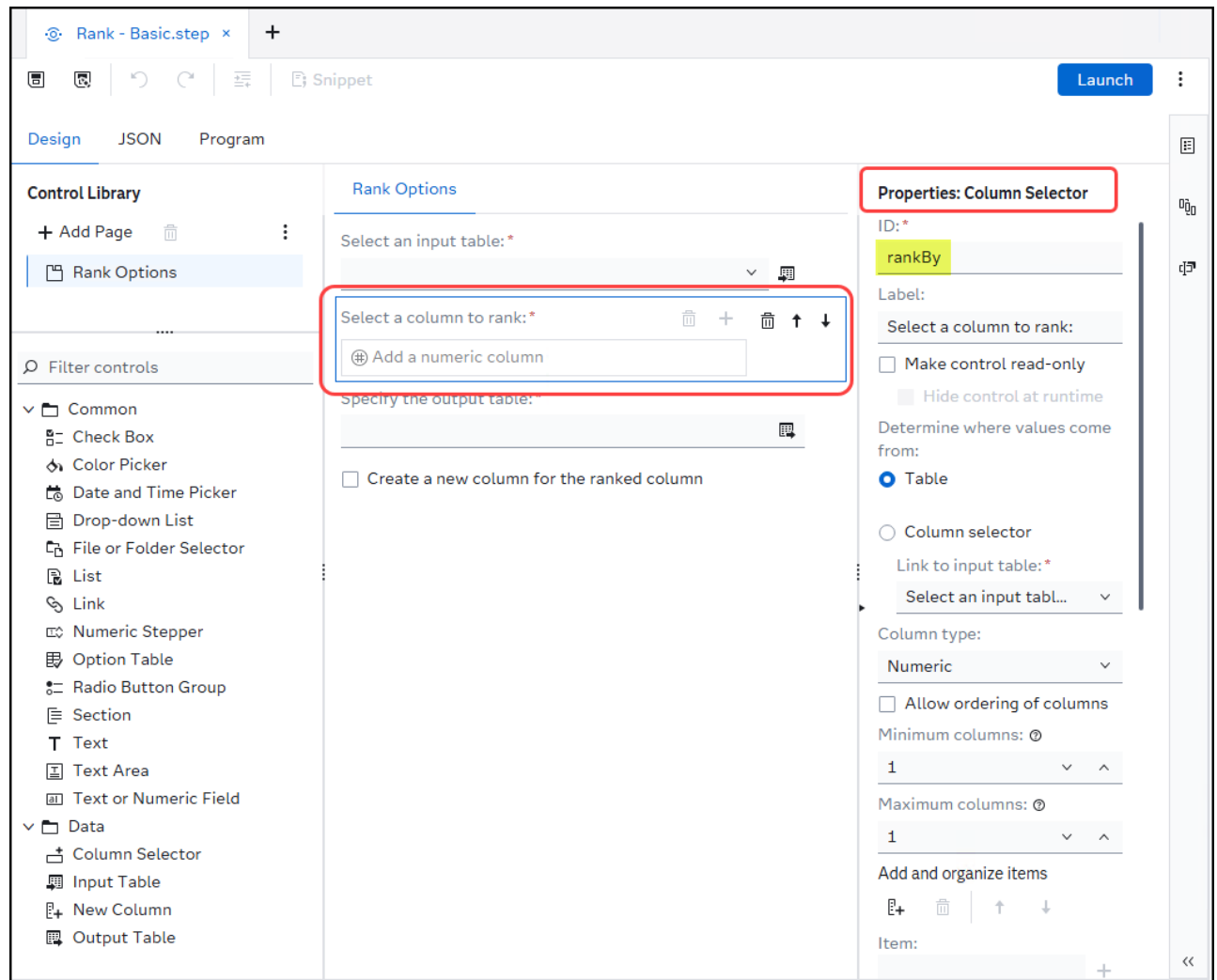
There are also more complex *starter template* examples to review; however, for this exercise you will focus on the basic example.

2. Select the **Select an input table** control to view its properties.



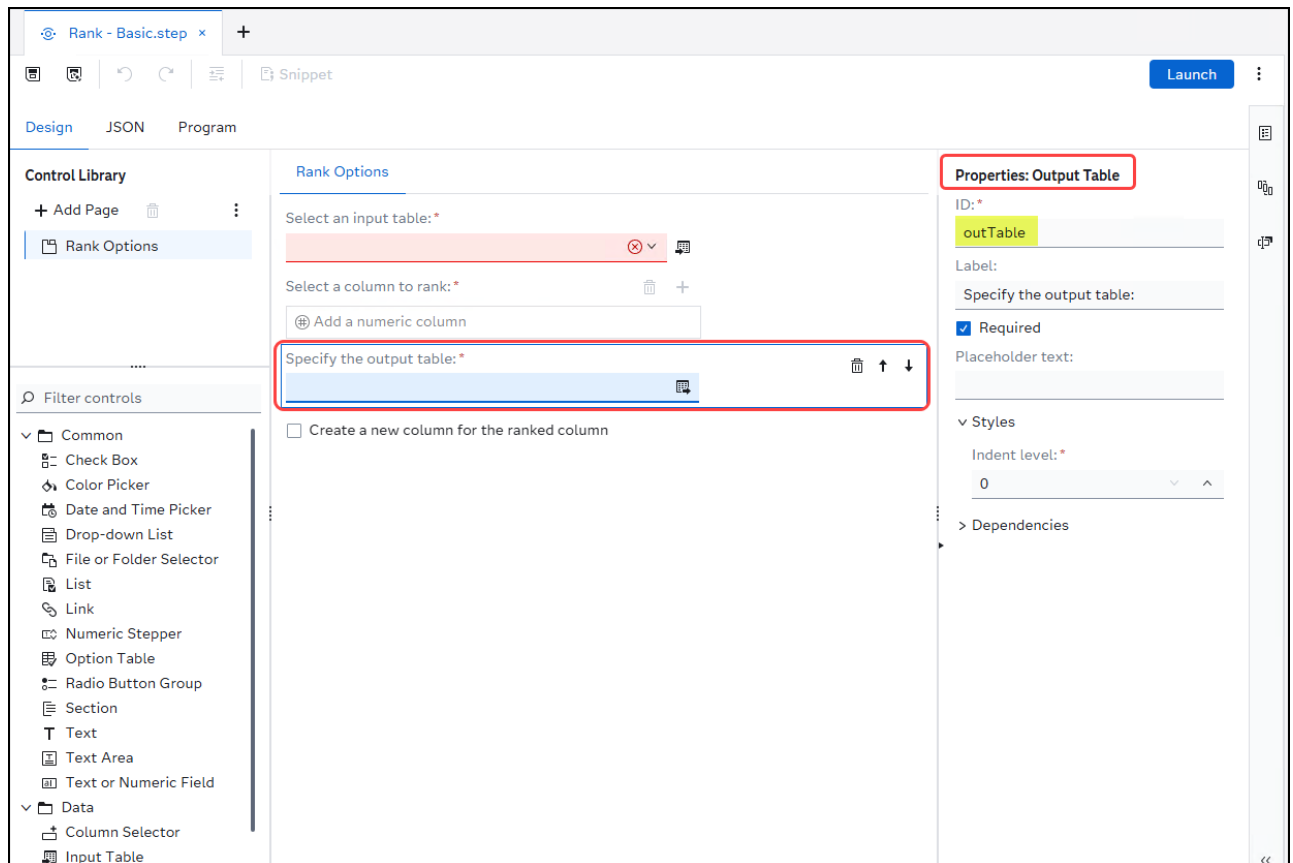
The property *ID* for the control is **inTable**.

3. Select the **Add a numeric column** control to view its properties.



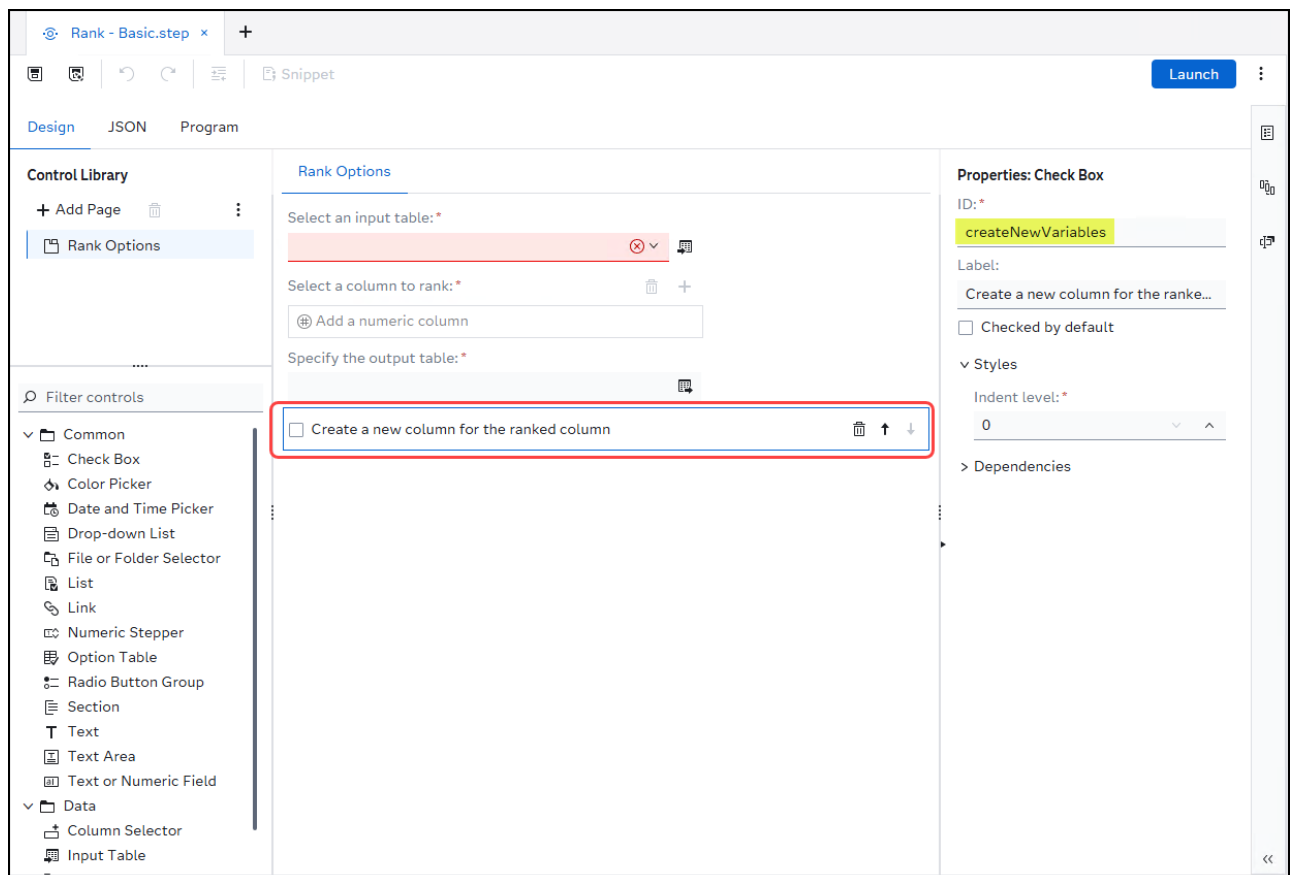
✎ The property *ID* for the control is **rankBy**.

4. Select the **Specify the output table** control to view its properties.



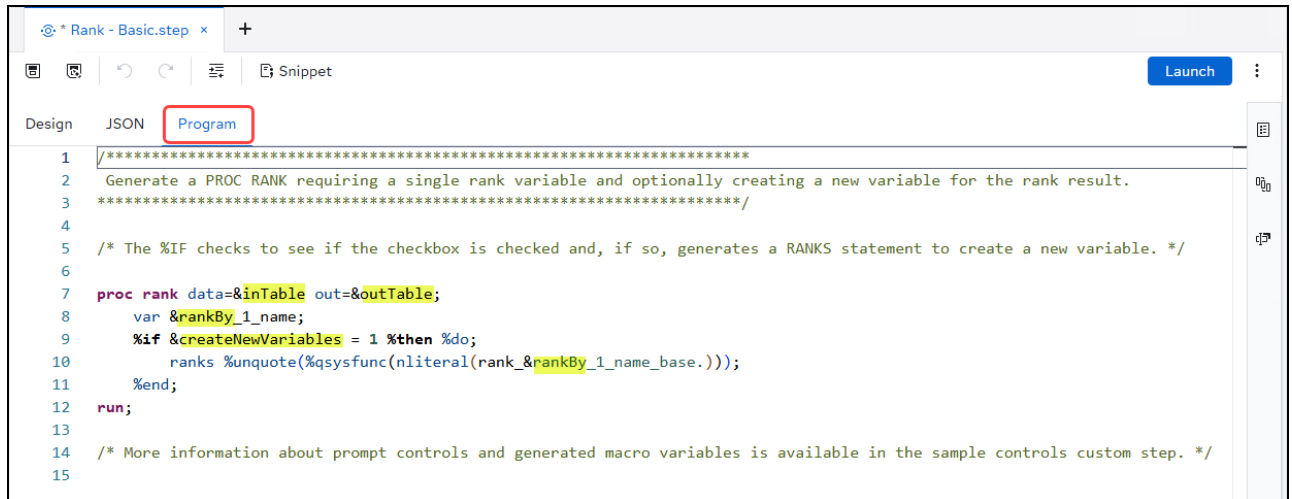
✎ The property *ID* for the control is **outTable**.

5. Select the **Create a new column for the ranked column** *check-box* control to view its properties.



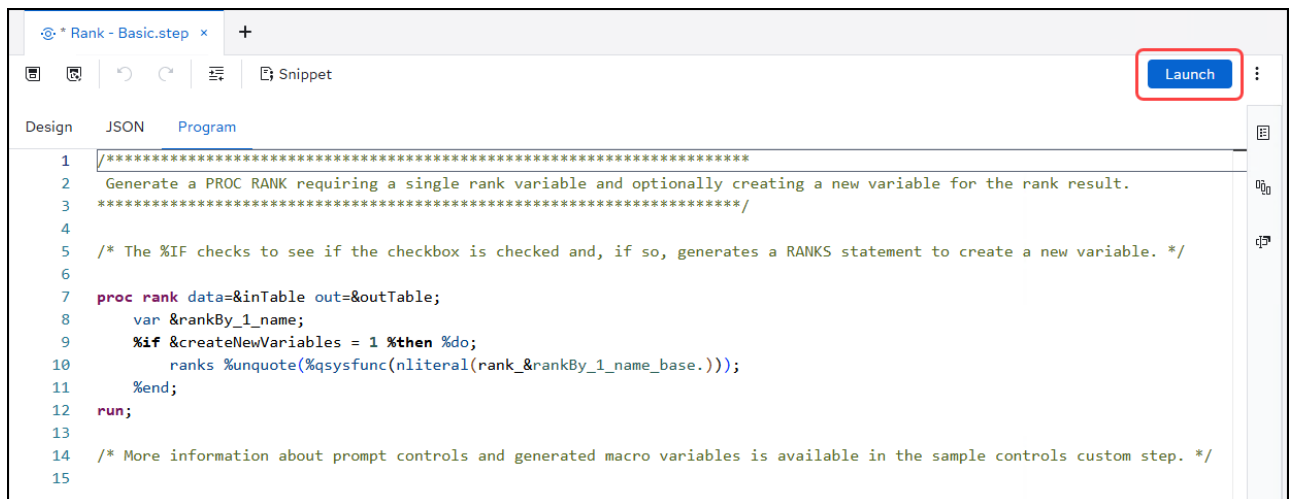
✎ The property *ID* for the control is **createNewVariables**.

6. Select the **Program** tab to view the code for the custom step.



✎ The property *ID* references for the various controls from the user interface design for the custom step.

7. Click **Launch** to test the sample custom step in stand-alone mode.



8. Make the following selections:

- Select an input table: **SASHELP.BASEBALL**.

✎ Click  to navigate to the desired input table.

- Select a column to rank: **CrHits**

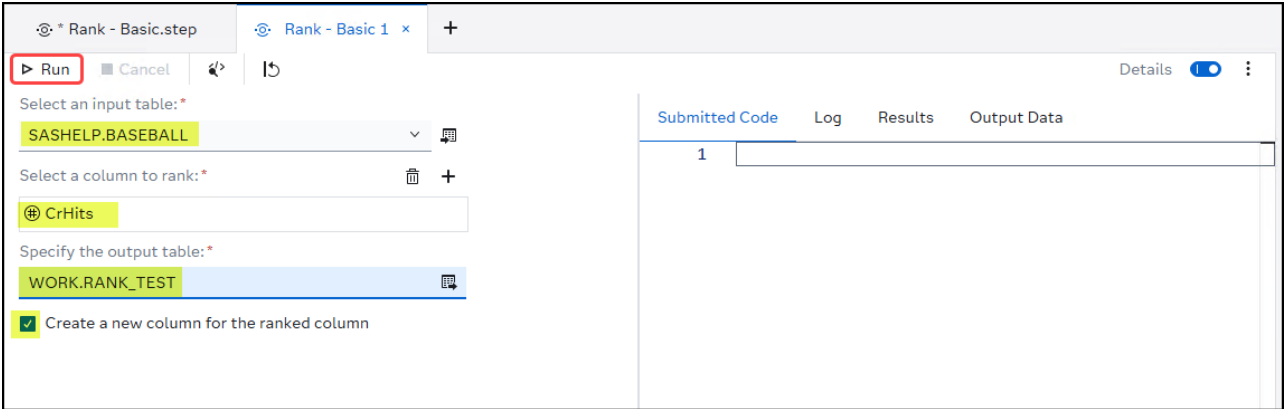
✎ Click  to select a numeric column from the input table.

- Specify an output table: **WORK.RANK_TEST**

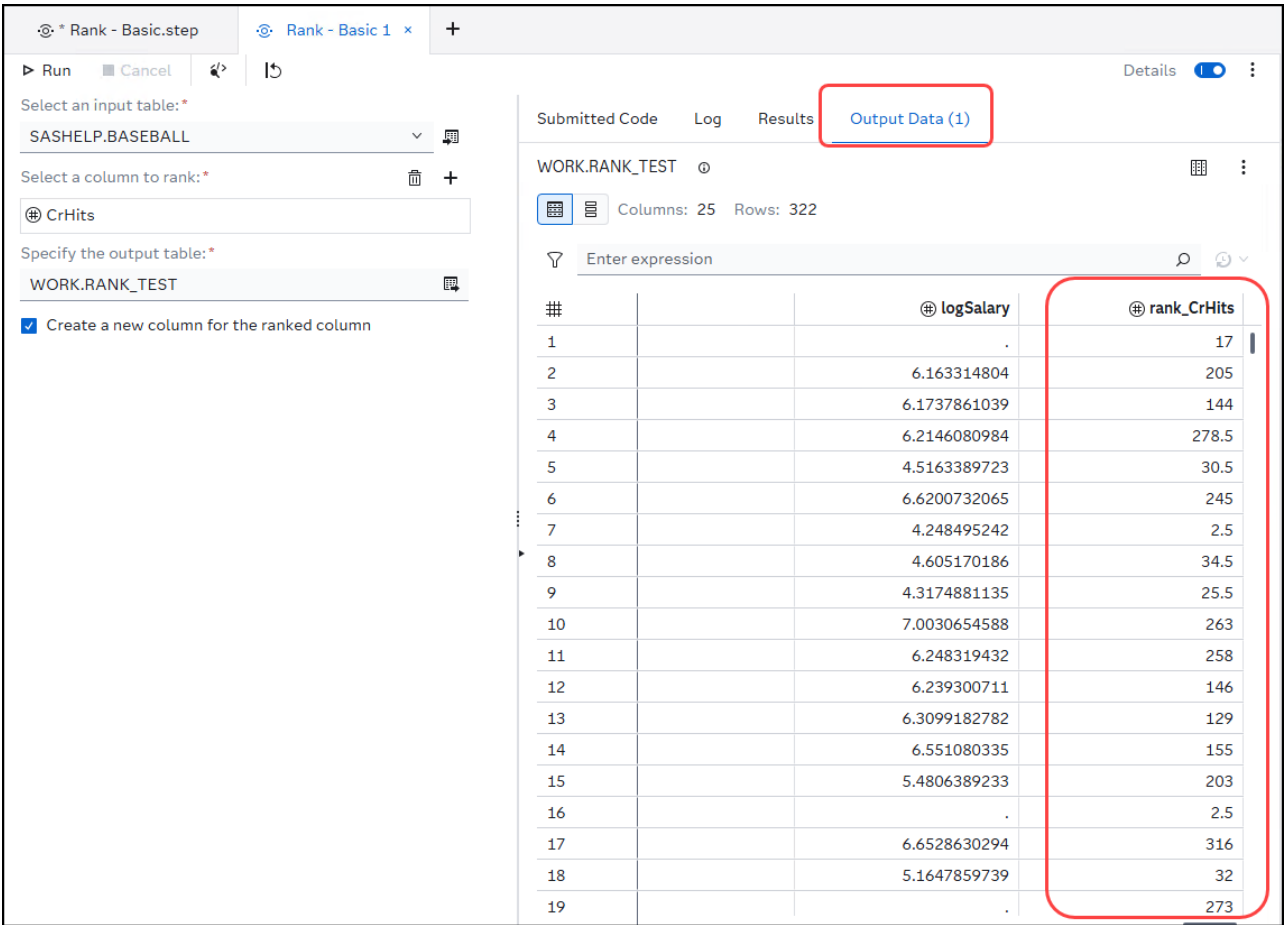
✎ Click  to navigate to the desired output library and then enter the desired output table name.

-  Create a new column for the ranked column

9. Click  to run the custom step.



10. Select the **Output Data (1)** tab to view the resulting output from the custom step.

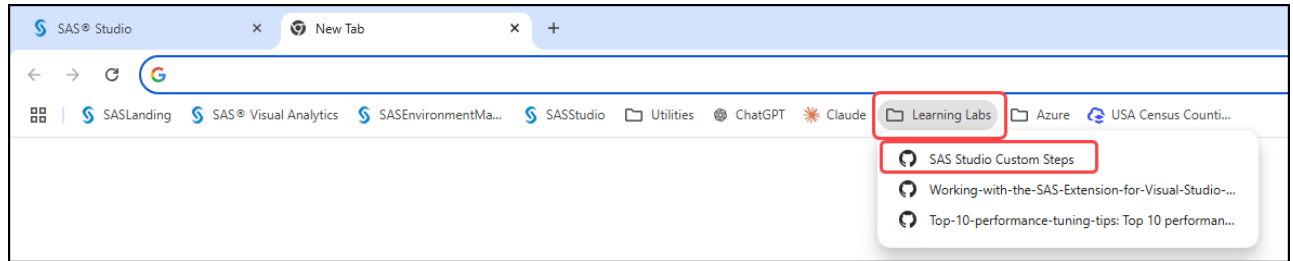


- 11. Click **x** to close the *Rank - Basic 1* tab.
- 12. Click **x** to close the *Rank-Basic.step* tab.
- 13. Select **Don't Save** if prompted when closing the tab.

3. Search the Custom Step GitHub Repository

- 1. Open a new tab in the *Google Chrome* browser.

2. Select **Learning Labs → SAS Studio Custom Steps** to open the SAS GitHub repository for custom steps.



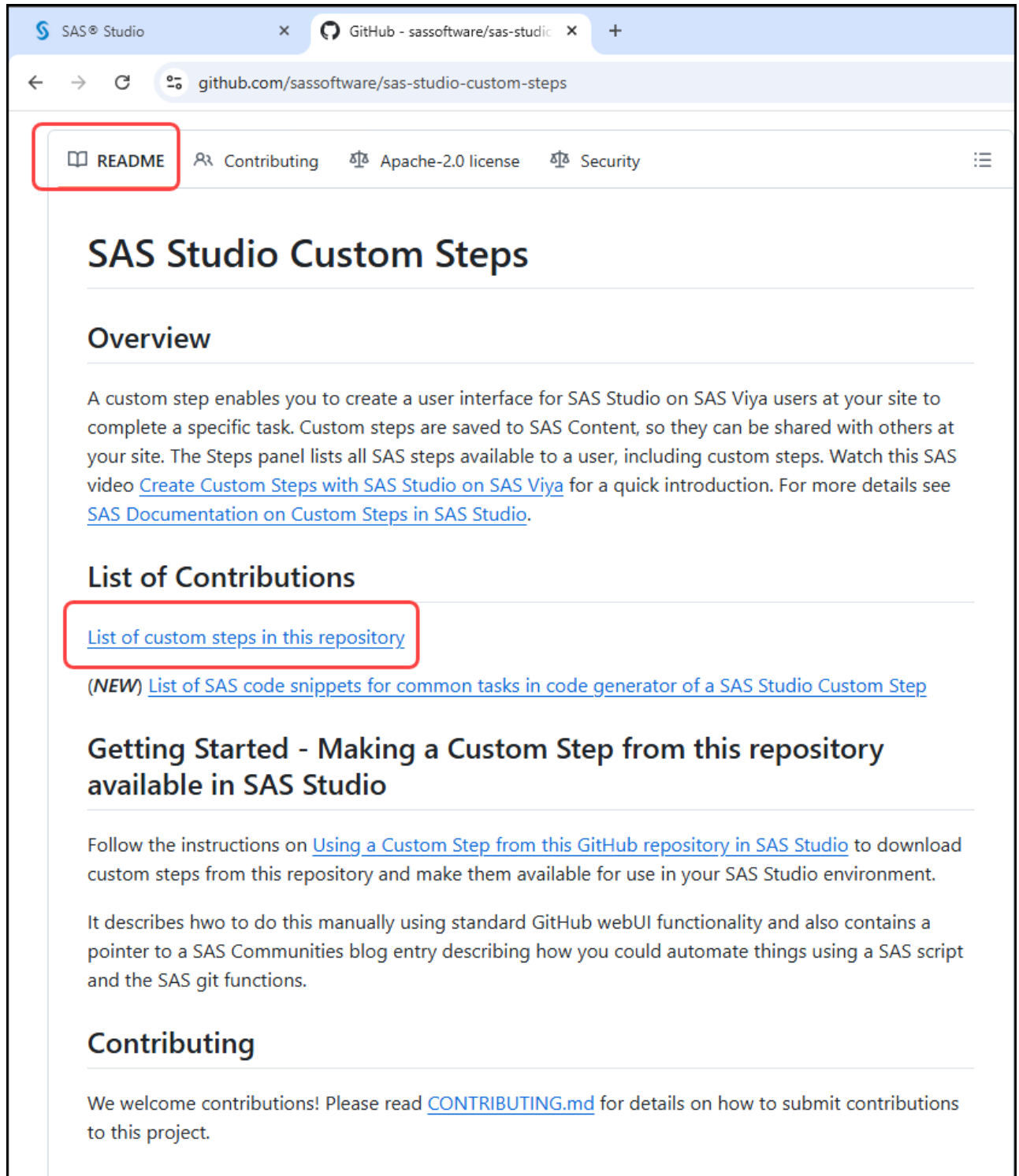
✎ This is a saved link to the SAS Studio Custom Steps GitHub repository:

<https://github.com/sassoftware/sas-studio-custom-steps>.

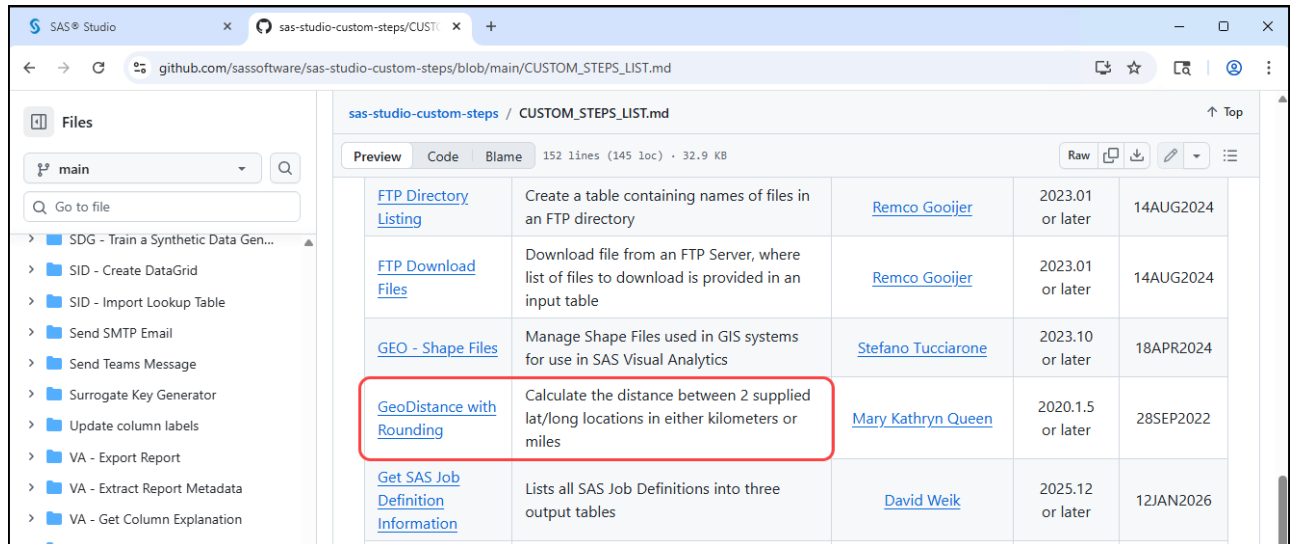
✎ This is a SAS Community article on how you can download all the custom steps for this repository:

<https://communities.sas.com/t5/SAS-Communities-Library/Uploading-all-Custom-Steps-from-the-SAS-Software-GitHub/ta-p/887903>

3. Click **Sign In** if prompted to sign in to the repository.
4. Scroll down towards to the bottom of the page to view the **README** information for the repository.
5. Select the link for **List of custom steps in this repository**.



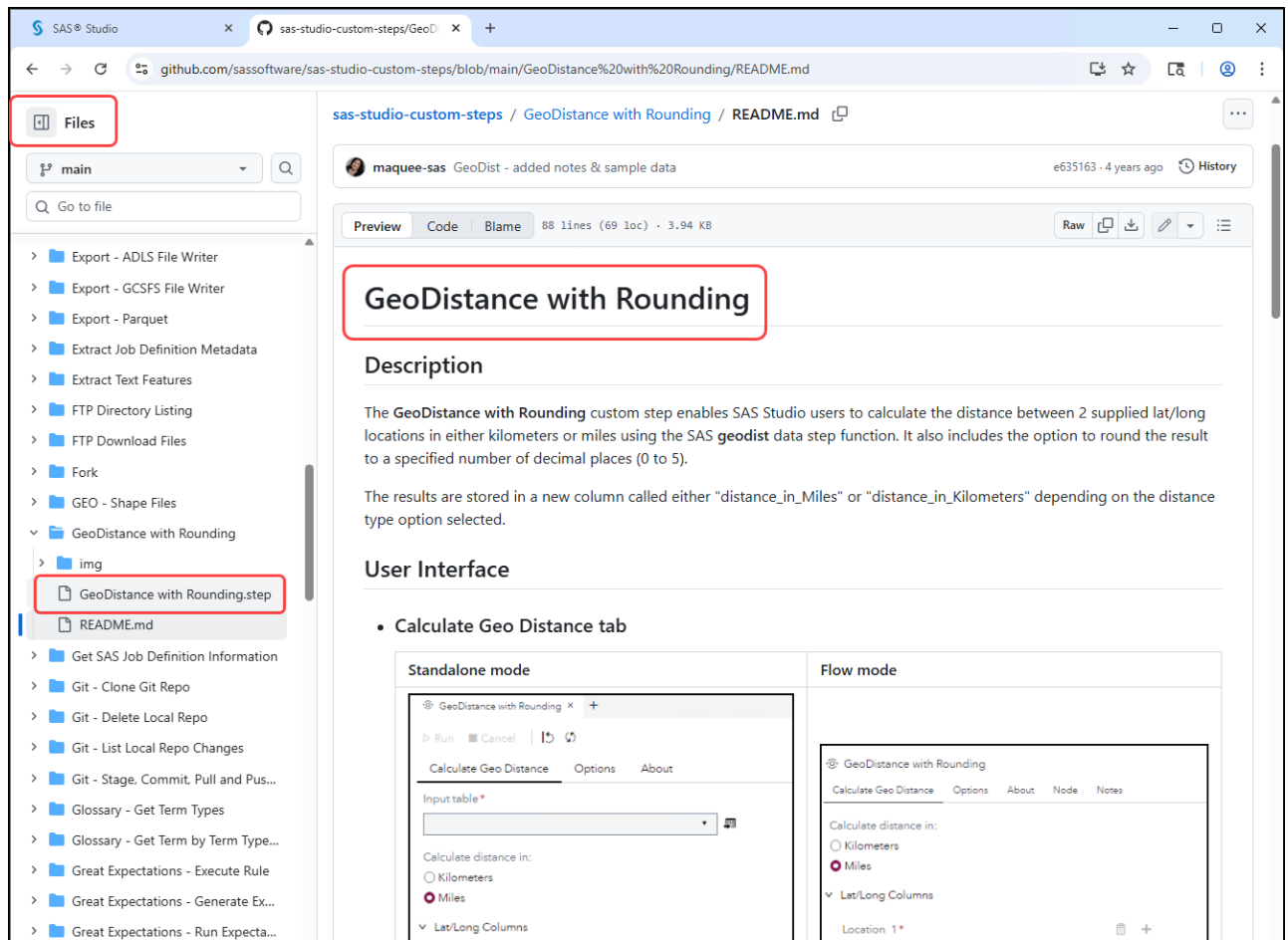
6. Scroll down until you locate the entry for **GeoDistance with Rounding** in the table of *Available Custom Steps*.



Step Name	Description	Author	Version	Date
FTP Directory Listing	Create a table containing names of files in an FTP directory	Remco Gooijer	2023.01 or later	14AUG2024
FTP Download Files	Download file from an FTP Server, where list of files to download is provided in an input table	Remco Gooijer	2023.01 or later	14AUG2024
GEO - Shape Files	Manage Shape Files used in GIS systems for use in SAS Visual Analytics	Stefano Tucciarone	2023.10 or later	18APR2024
GeoDistance with Rounding	Calculate the distance between 2 supplied lat/long locations in either kilometers or miles	Mary Kathryn Queen	2020.1.5 or later	28SEP2022
Get SAS Job Definition Information	Lists all SAS Job Definitions into three output tables	David Weik	2025.12 or later	12JAN2026

7. Select the link for **GeoDistance with Rounding** to view additional descriptive information about the custom step.

8. On the left-hand side in the *Files* section, select the **GeoDistance with Rounding.step**.



GeoDistance with Rounding

Description

The **GeoDistance with Rounding** custom step enables SAS Studio users to calculate the distance between 2 supplied lat/long locations in either kilometers or miles using the SAS **geodist** data step function. It also includes the option to round the result to a specified number of decimal places (0 to 5).

The results are stored in a new column called either "distance_in_Miles" or "distance_in_Kilometers" depending on the distance type option selected.

User Interface

- Calculate Geo Distance tab

Standalone mode

GeoDistance with Rounding

Calculate Geo Distance Options About

Input table *

Calculate distance in:

☐ Kilometers

☒ Miles

Lat/Long Columns

Location 1 *

Flow mode

GeoDistance with Rounding

Calculate Geo Distance Options About Node Notes

Calculate distance in:

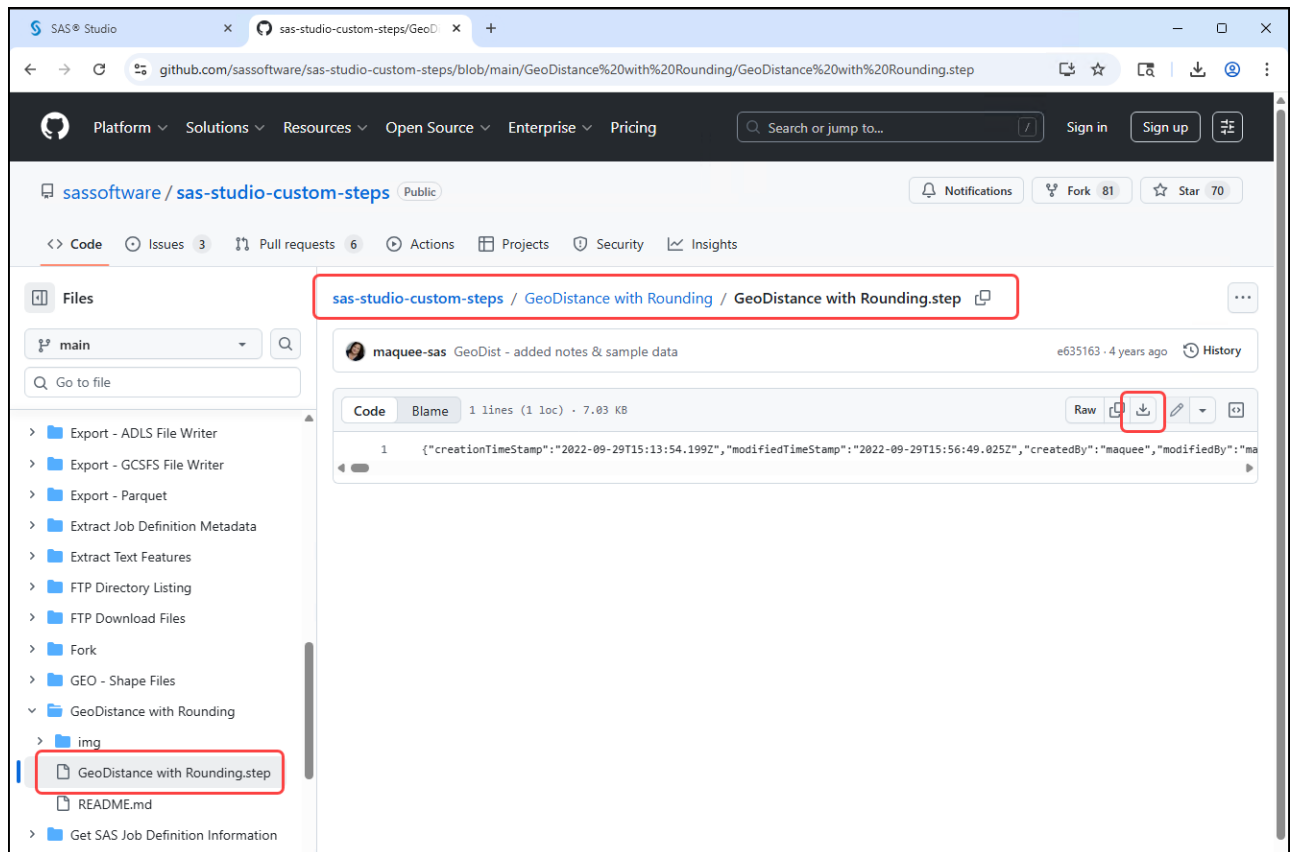
☐ Kilometers

☒ Miles

Lat/Long Columns

Location 1 *

9. Click  to download the file for this custom step.

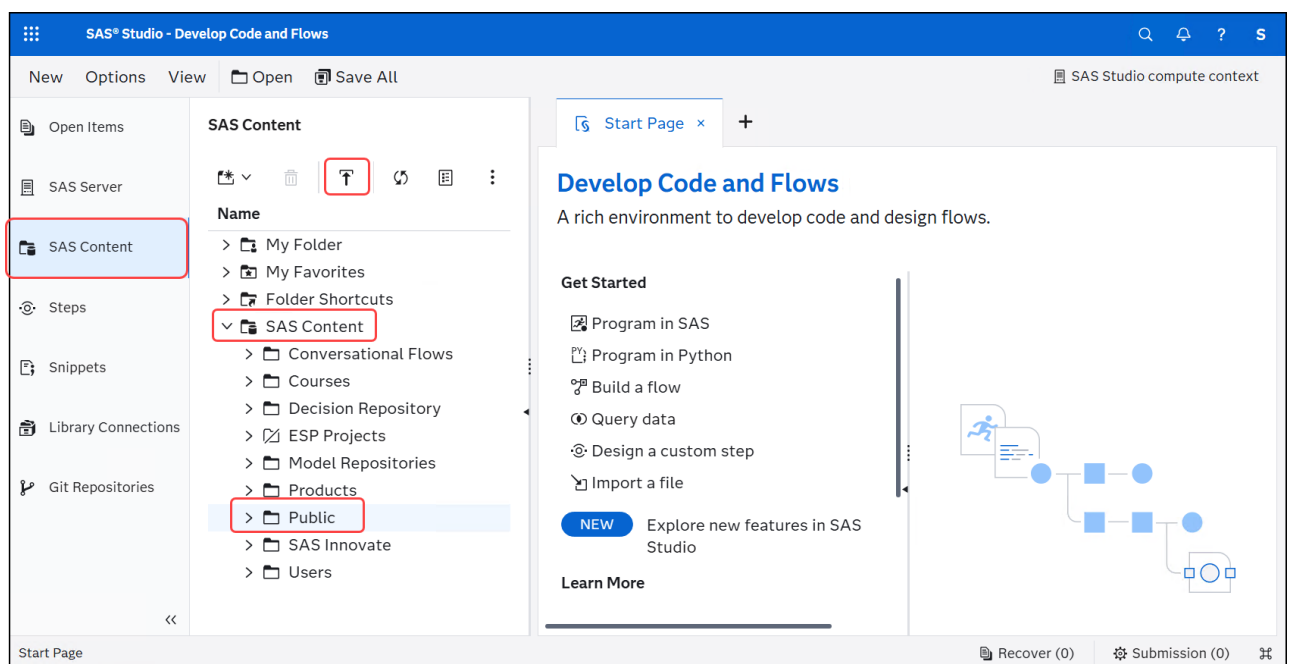


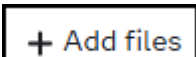
10. Return to the **SAS Studio** tab in the *Google Chrome* browser.

11. Select  to view the *SAS Content* folders.

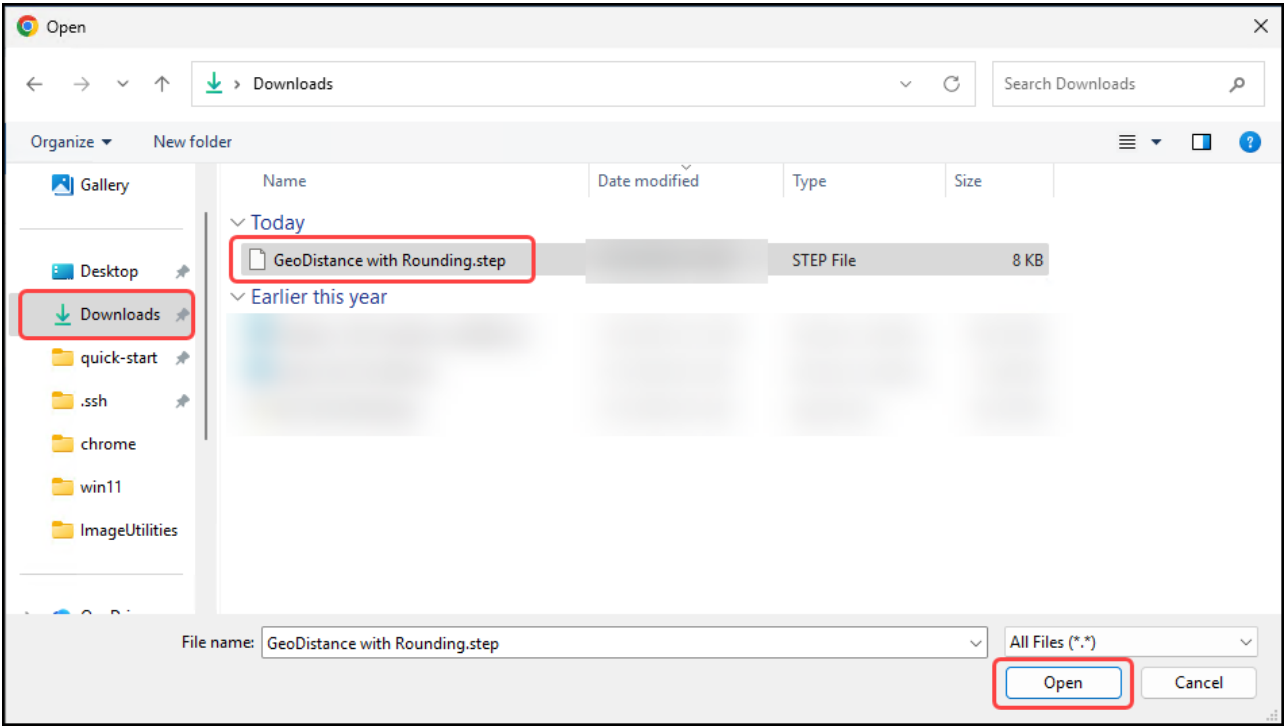
12. Expand the *SAS Content* folder and select the **Public** folder.

13. Click  to upload files to the selected folder.

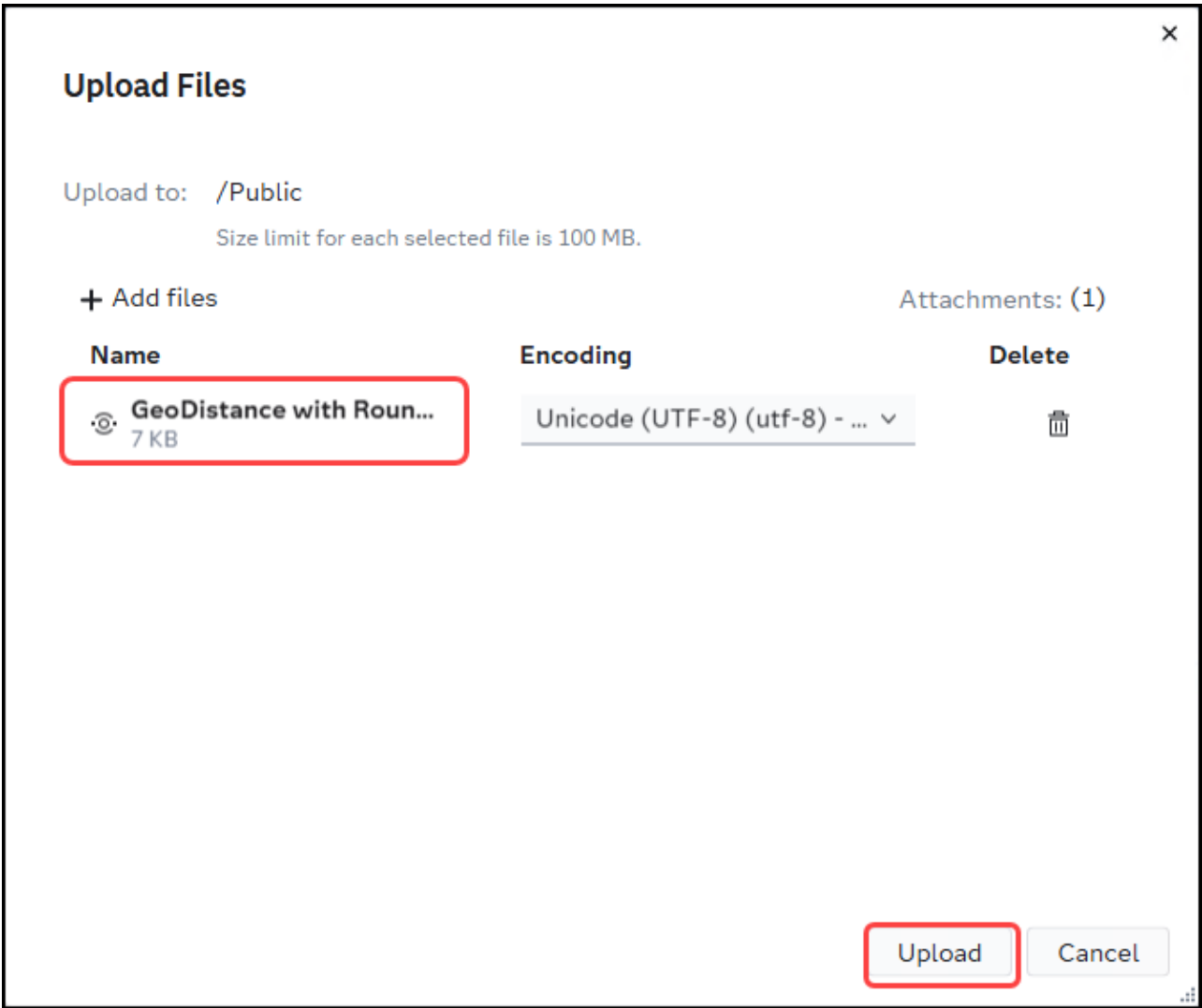


14. Click  and navigate to **Downloads** folder.

15. Select the **GeoDistance with Rounding.step** file and click **Open**.

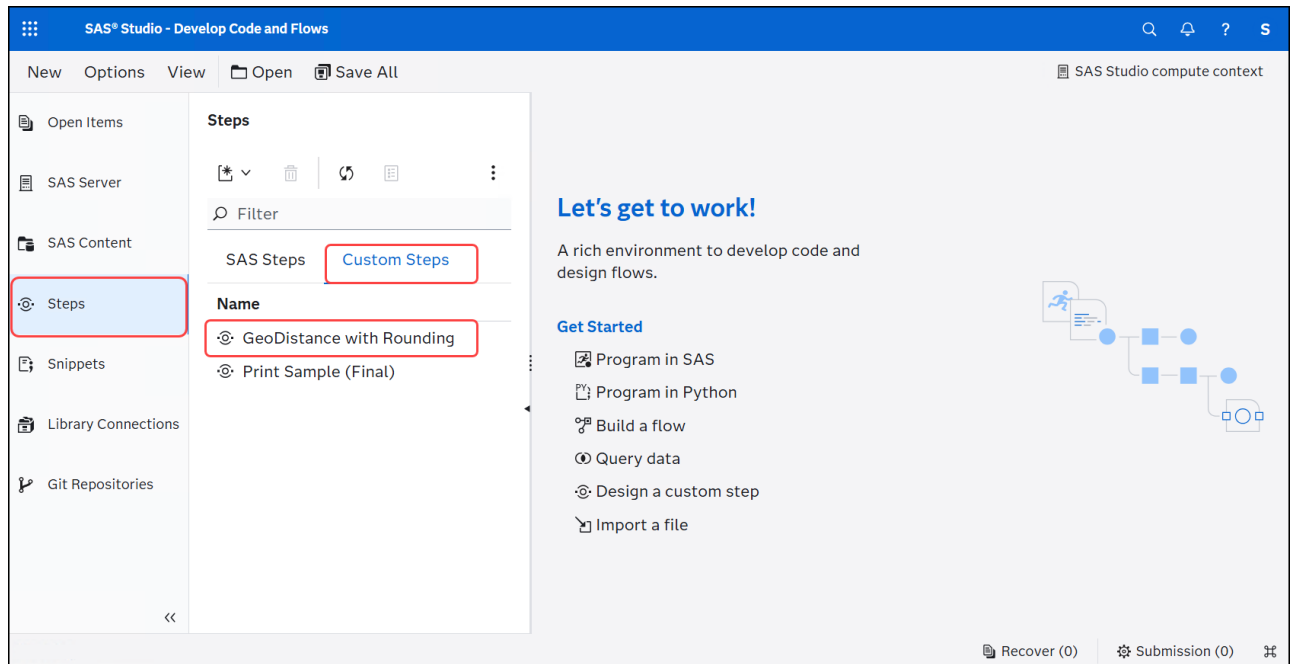


16. Click **Upload** to upload the attached file(s).



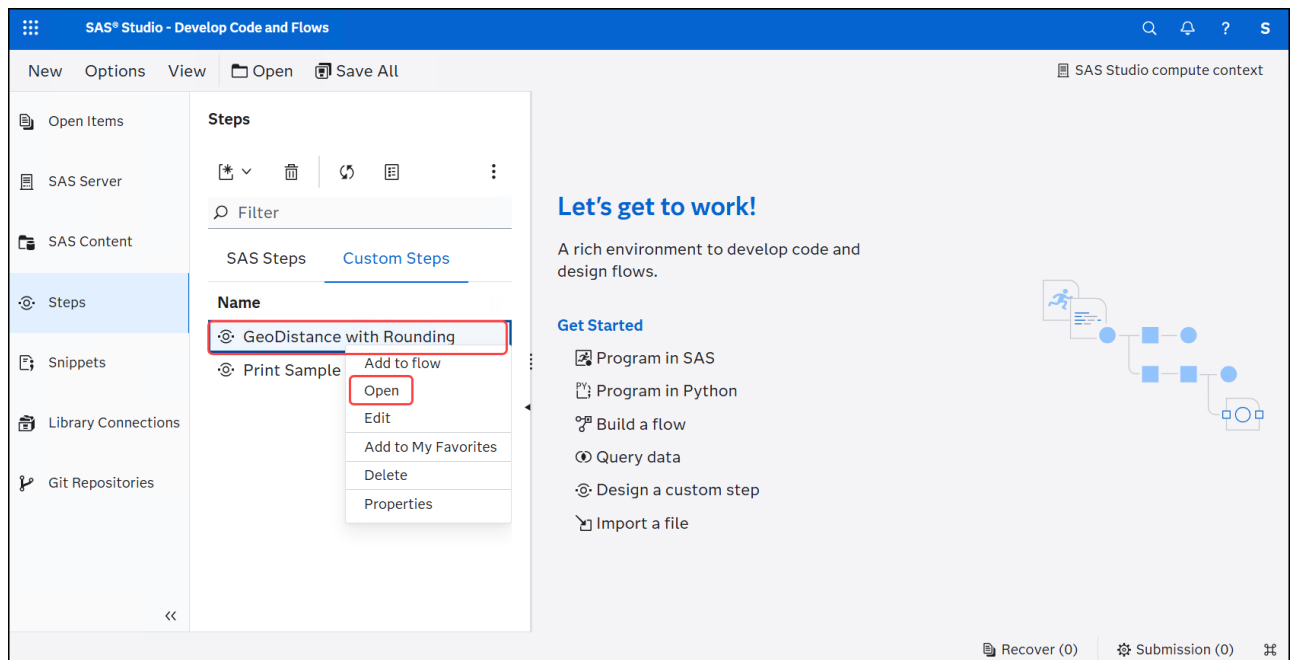
17. Select  to view the **Steps** pane.

18. Select the **Custom Steps** to confirm the *GeoDistance with Rounding* custom step is listed.



4. Test the Custom Step using Stand-alone Mode

1. In the *Steps* pane, select the **GeoDistance with Rounding** custom step and right-click then select **Open** to open the custom step in stand-alone mode.

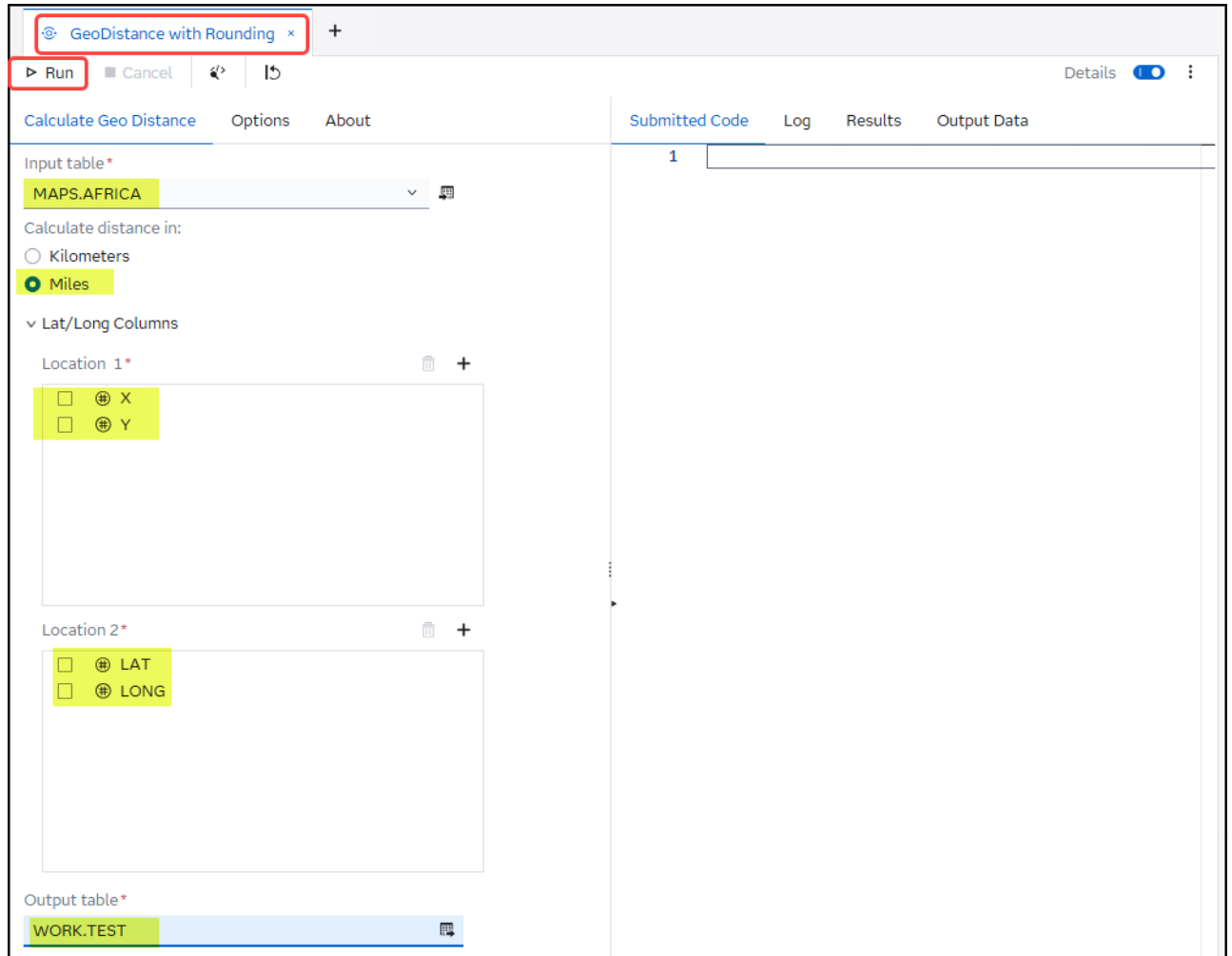


2. Select the following required information for the custom step:

- Input table: **MAPS.AFRICA**
- Calculate distance in: **Miles**
- Location 1:

- **X**
- **Y**
- Location 2:
 - **LAT**
 - **LONG**
- Output table: **WORK.TEST**

3. Click  **Run** to run the custom step in stand-alone mode.



The screenshot shows the configuration window for the 'GeoDistance with Rounding' step. The window has a title bar with the step name and a close button. Below the title bar is a toolbar with 'Run', 'Cancel', and other icons. The main area is divided into two panes. The left pane contains the configuration options, and the right pane shows the 'Submitted Code' tab with a table containing one row.

Configuration Options:

- Input table:** MAPS.AFRICA
- Calculate distance in:** ☒ Miles
- Lat/Long Columns:**
 - Location 1:** ☒ X, ☒ Y
 - Location 2:** ☒ LAT, ☒ LONG
- Output table:** WORK.TEST

Submitted Code Tab:

Submitted Code
1

4. Review the **Output Data** tab.

GeoDistance with Rounding

RunCancel

Calculate Geo DistanceOptionsAbout

Input table*
MAPS.AFRICA

Calculate distance in:

Kilometers

Miles

Lat/Long Columns

Location 1*

X

Y

Location 2*

LAT

LONG

Output table*
WORK.TEST

Submitted CodeLogResultsOutput Data (1)

WORK.TEST

Columns: 9Rows: 52,824

Enter expression

#	LAT	LONG	distance_in_Miles
1	0.4763100675	0.1512618584	65.472588797
2	0.4665992501	0.1342788354	63.89790514
3	0.4560400078	0.1162873997	62.217721563
4	0.4472357814	0.1016120945	60.839560605
5	0.4363323078	0.083874638	59.162371164
6	0.4288467945	0.0719608896	58.028930644
7	0.4214776259	0.0604368686	56.927055882
8	0.4099293542	0.0427314778	55.225587043
9	0.4018329513	0.0305529678	54.050067918
10	0.3946674438	0.01994523	53.021995608
11	0.3866631509	0.0082563867	51.886056752
12	0.3810150668	0.0001405863	51.093599576
13	0.3683080954	-0.020415509	49.240100017
14	0.3675275593	-0.020691838	49.162707119
15	0.3665627803	-0.020895479	49.072022632
16	0.366048863	-0.020725785	49.033786438
17	0.3654283163	-0.020813046	48.977050129
18	0.3647834953	-0.020706392	48.925261187
19	0.3631496969	-0.02025552	48.80073741
20	0.3623497447	-0.02027006	48.731234473
21	0.3620006646	-0.020376714	48.697239119
22	0.36185523	-0.020575483	48.677430598

5. Review the Log for Macro Variable Names and Values

1. Select the **Log** tab.
2. Expand the section named **/*region: Generated macro initialization*/**.

The screenshot shows the SAS Studio interface for a custom step named 'GeoDistance with Rounding'. The 'Log' tab is active, showing the following SAS macro code:

```

>1  /* region: Generated preamble */...
79
>80 /* region: Generated macro initialization */...
202 /* sas code provided in template section of custom Step - BEGIN */
203 /* Set Input and Output Data */
204 data &outTable ;
205 set &inTable ;
206
207
208 /* newColumn is a hidden control in the UI. The code sets the name of the new output
209
210 /* Calculate GeoDistance in Miles */
211 %if &disType=Miles %then %do ;
212 %let newColumn_name=distance_in_Miles ;
213 &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'M')
214 %end;
215
216 /* Calculate GeoDistance in Kilometers */
217 %else %do ;
218 %let newColumn_name=distance_in_Kilometers ;
219 &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'K')
220 %end ;
221
222
223 /* Round GeoDistance Value */
224 %if &round=1 %then %do ;
225 select(&decimalPlaces) ;
226 when(0) &newColumn_name=round(distance_in_&disType) ;
227 when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
228 when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
229 when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
230 when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
231 when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
232 end;
233 %end;
234 run;

```

The red box highlights the section heading **/* region: Generated macro initialization */...** at line 80.

3. Scroll up to the section heading named **/* Macro variables derived from user input to this step - BEGIN */** to view macro variable names and current values for the user input from the custom step.

The screenshot shows the SAS Studio interface for the same custom step. The 'Log' tab is active, showing the following SAS macro code:

```

103
104 %end;
105 %mend;
106
107 /* Define utility macros - END */
108
109 /* Macro variables derived from user input to this step - BEGIN */
110
111 /* Macro variable(s) for UI control with ID of decimalPlaces */
112 %let decimalPlaces=;
113
114 /* Macro variable(s) for UI control with ID of disType */
115 %let disType=Miles;
116
117 /* Macro variable(s) for UI control with ID of inTable */
118 %let inTable=MAPS.AFRICA;
119 %let inTable_eTag=%nrquote("%jqpvqm4o%");
120 %let inTable_engine=V9;
121 %let inTable_id=AFRICA;
122 %let inTable_isCas=false;
123 %let inTable_label=%nrquote("AFRICA: Source - Derived from CIA 2001 World Shapefile -
124 %let inTable_lib=MAPS;
125 %let inTable_libref=MAPS;
126 %let inTable_name=AFRICA;
127 %let inTable_name_base=AFRICA;
128 %let inTable_tblType=table;
129 %let inTable_type=dataTable;
130
131 /* Macro variable(s) for UI control with ID of loc1 */
132 %let loc1=X Y;
133 %let loc1_count=2;
134
135 /* Macro variables for column <X> */
136 %let loc1_1_format=;
137 %let loc1_1_informat=;
138 %let loc1_1_label=%nrquote("Projected Longitude: Albers");
139 %let loc1_1_name=X;

```

The red box highlights the section heading **/* Macro variables derived from user input to this step - BEGIN */** at line 109.

4. Notice that the macro variable **disType** is set to **Miles**.

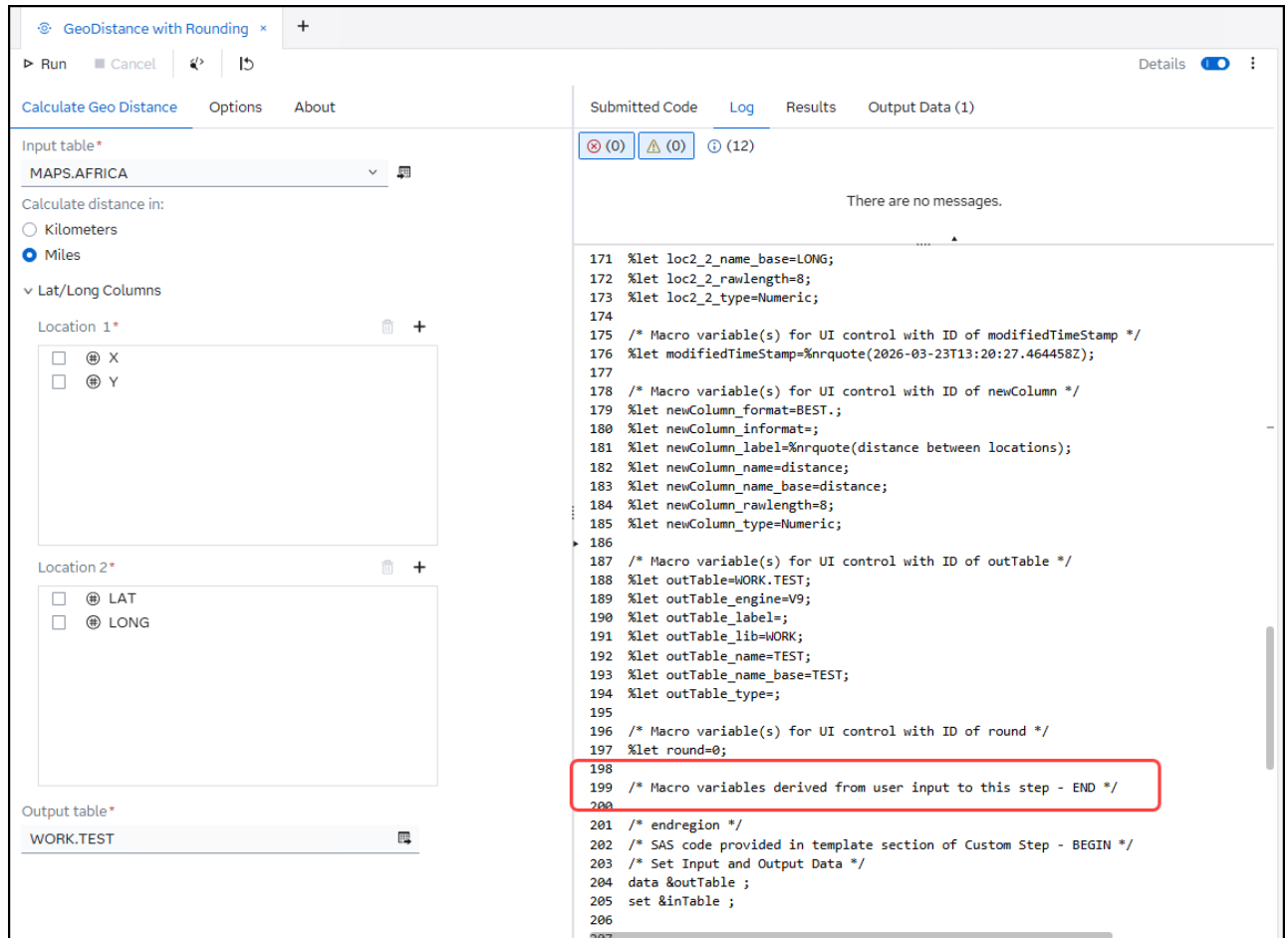
The screenshot displays the 'GeoDistance with Rounding' custom step configuration. On the left, the 'Input table' is set to 'MAPS.AFRICA'. Under 'Calculate distance in:', 'Miles' is selected. Under 'Lat/Long Columns', 'Location 1' is selected with both 'X' and 'Y' coordinates. The 'Output table' is 'WORK.TEST'. On the right, the 'Submitted Code' pane shows a list of macro variables. Lines 114 and 115 are highlighted with a red box:

```

114 /* Macro variable(s) for UI control with ID of disType */
115 %let disType=Miles;

```

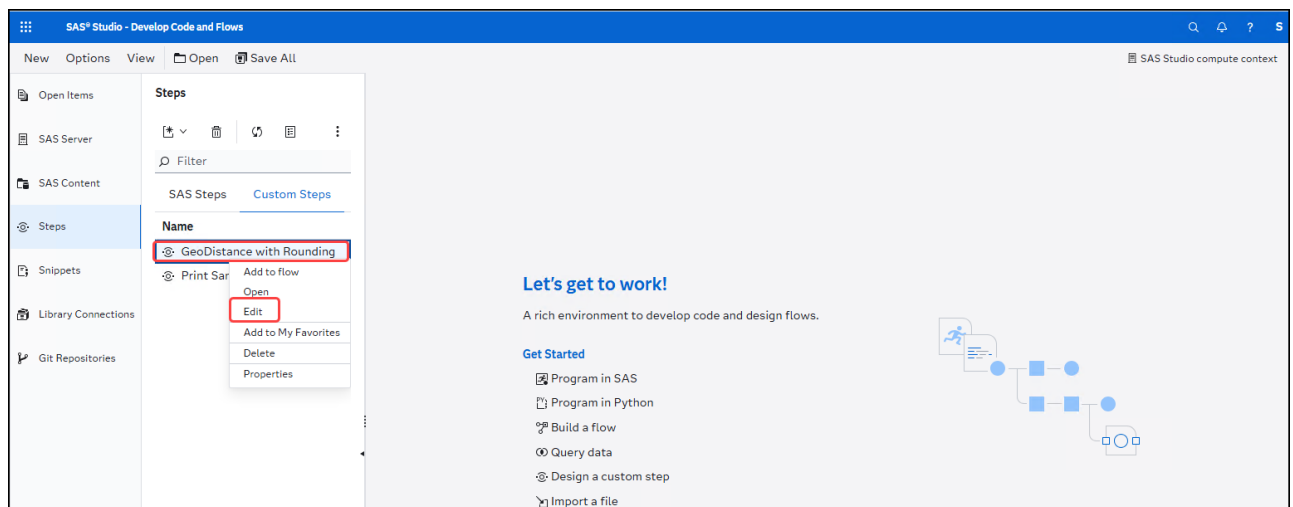
5. Scroll to the section closer named **/* Macro variables derived from user input to this step - END */**. This ends the section of the macro variable names and values for the user input from the custom step.



6. Click **x** to close the open custom step in stand-alone mode.

6. Use %put Statements to Aid in the Testing of your Code

1. In the *Steps* pane, select the **GeoDistance with Rounding** custom step and right-click then select **Edit** to open the custom step in edit mode.



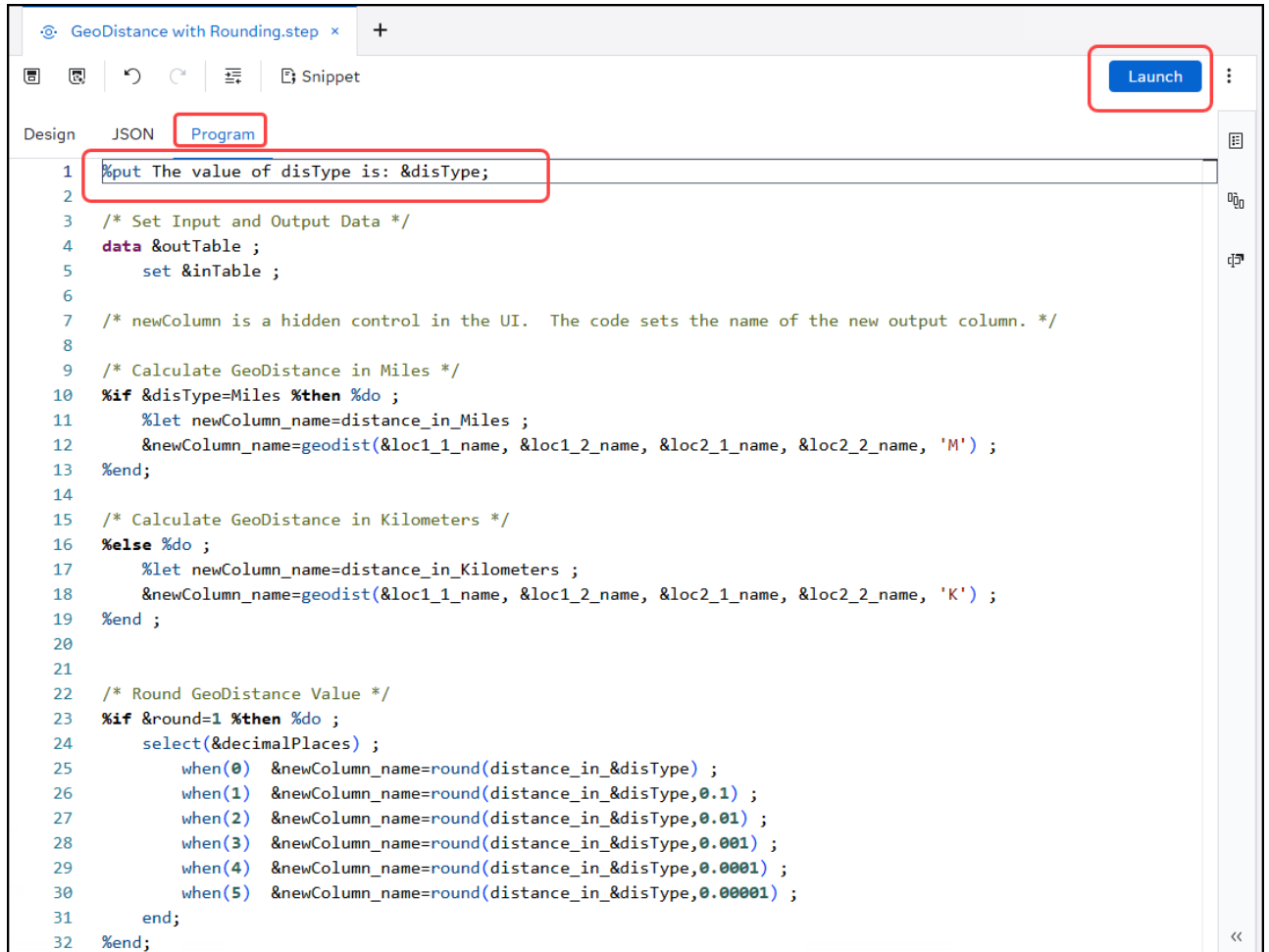
2. Select the **Program** tab to view the code for the custom step.

3. Add the following line at the beginning of the code block:

```
%put The value of disType is: &disType;
```

4. Click  to save the changes to the program code.

5. Click **Launch** to test the custom step.



```

1 %put The value of disType is: &disType;
2
3 /* Set Input and Output Data */
4 data &outTable ;
5   set &inTable ;
6
7 /* newColumn is a hidden control in the UI. The code sets the name of the new output column. */
8
9 /* Calculate GeoDistance in Miles */
10 %if &disType=Miles %then %do ;
11   %let newColumn_name=distance_in_Miles ;
12   &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'M') ;
13 %end;
14
15 /* Calculate GeoDistance in Kilometers */
16 %else %do ;
17   %let newColumn_name=distance_in_Kilometers ;
18   &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'K') ;
19 %end ;
20
21
22 /* Round GeoDistance Value */
23 %if &round=1 %then %do ;
24   select(&decimalPlaces) ;
25     when(0) &newColumn_name=round(distance_in_&disType) ;
26     when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
27     when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
28     when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
29     when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
30     when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
31   end;
32 %end;

```

6. Select the following required information to test the custom step:

- Input table: **MAPS.AFRICA**
- Calculate distance in: **Miles**
- Location 1:
 - **X**
 - **Y**
- Location 2:
 - **LAT**
 - **LONG**
- Output table: **WORK.TEST**

7. Click  to run the custom step in stand-alone mode.

8. Review the **Log** tab and scroll up to find the **%put The value of disType is: &disType;** statement.

GeoDistance with Rounding.step

Run Cancel

Calculate Geo Distance Options About

Input table*
MAPS.AFRICA

Calculate distance in:
☐ Kilometers
☒ Miles

Lat/Long Columns

Location 1*
☐ X
☐ Y

Location 2*
☐ LAT
☐ LONG

Output table*
WORK.TEST

Submitted Code Log Results Output Data (1)

There are no messages.

```

>1 /* region: Generated preamble */...
79
>80 /* region: Generated macro initialization */...
199 /* SAS code provided in template section of Custom Step - BEGIN */
200 %put The value of disType is: &disType;
    The value of disType is: Miles
201
202 /* Set Input and Output Data */
203 data &outTable ;
204 set &inTable ;
205
206 /* newColumn is a hidden control in the UI. The code sets the name of the new output column
207
208 /* Calculate GeoDistance in Miles */
209 %if &disType=Miles %then %do ;
210 %let newColumn_name=distance_in_Miles ;
211 %newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'M') ;
212 %end;
213
214 /* Calculate GeoDistance in Kilometers */
215 %else %do ;
216 %let newColumn_name=distance_in_Kilometers ;
217 %newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'K') ;
218 %end ;
219
220
221 /* Round GeoDistance Value */
222 %if &round=1 %then %do ;
223 select(&decimalPlaces) ;
224 when(0) &newColumn_name=round(distance_in_&disType) ;
225 when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
226 when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
227 when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
228 when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
229 when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
230 end;
  
```

9. Click **x** to close the *GeoDistance with Rounding 1* tab.

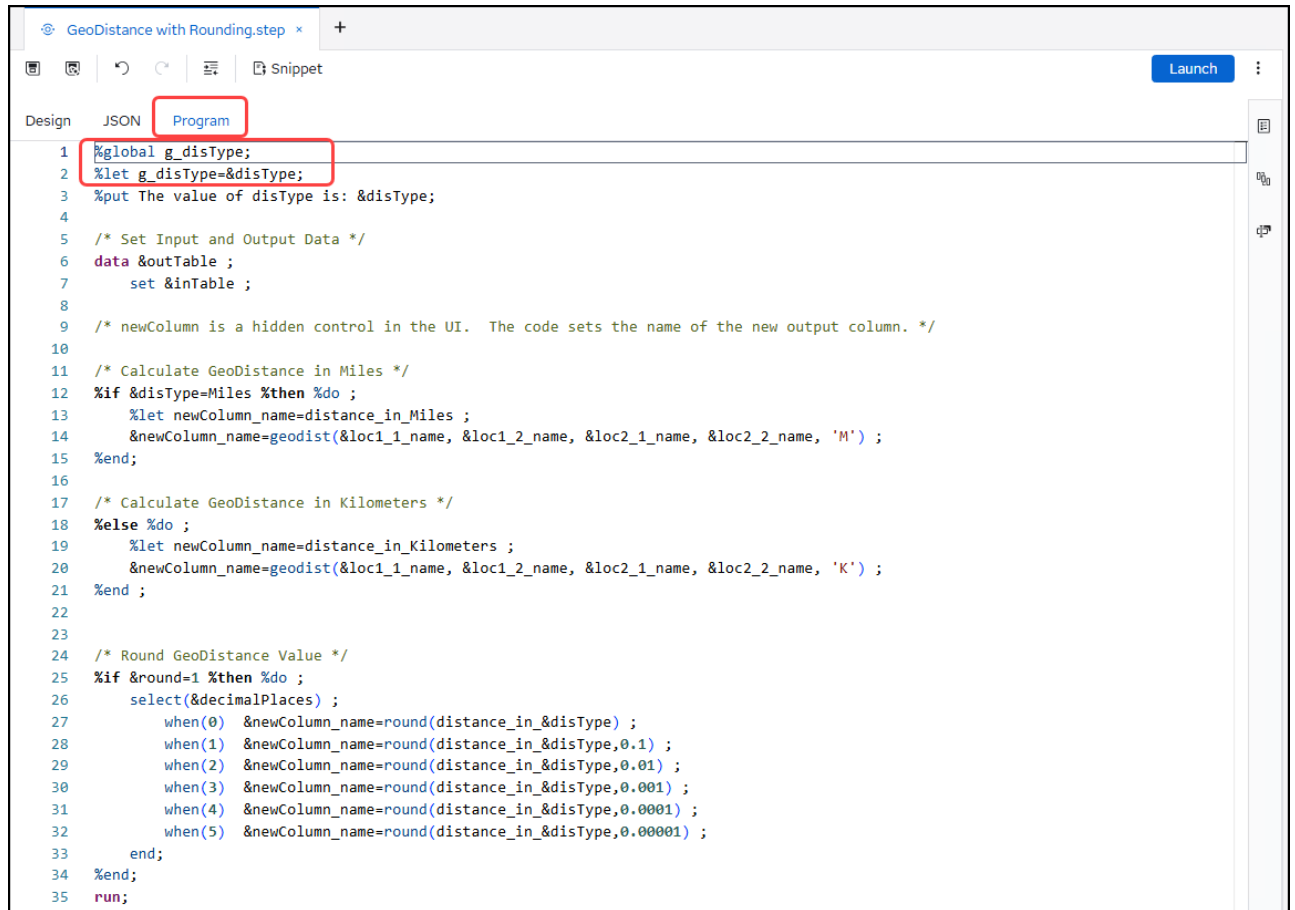
7. Use **%global** Statement to Assign Macro Variable Values for Use Outside of the Custom Step

1. On the **Program** tab, add the following lines of code at the beginning of the code block:

```

%global g_disType;
%let g_disType=&disType;
  
```

2. Click  to save the changes to the program code.



```

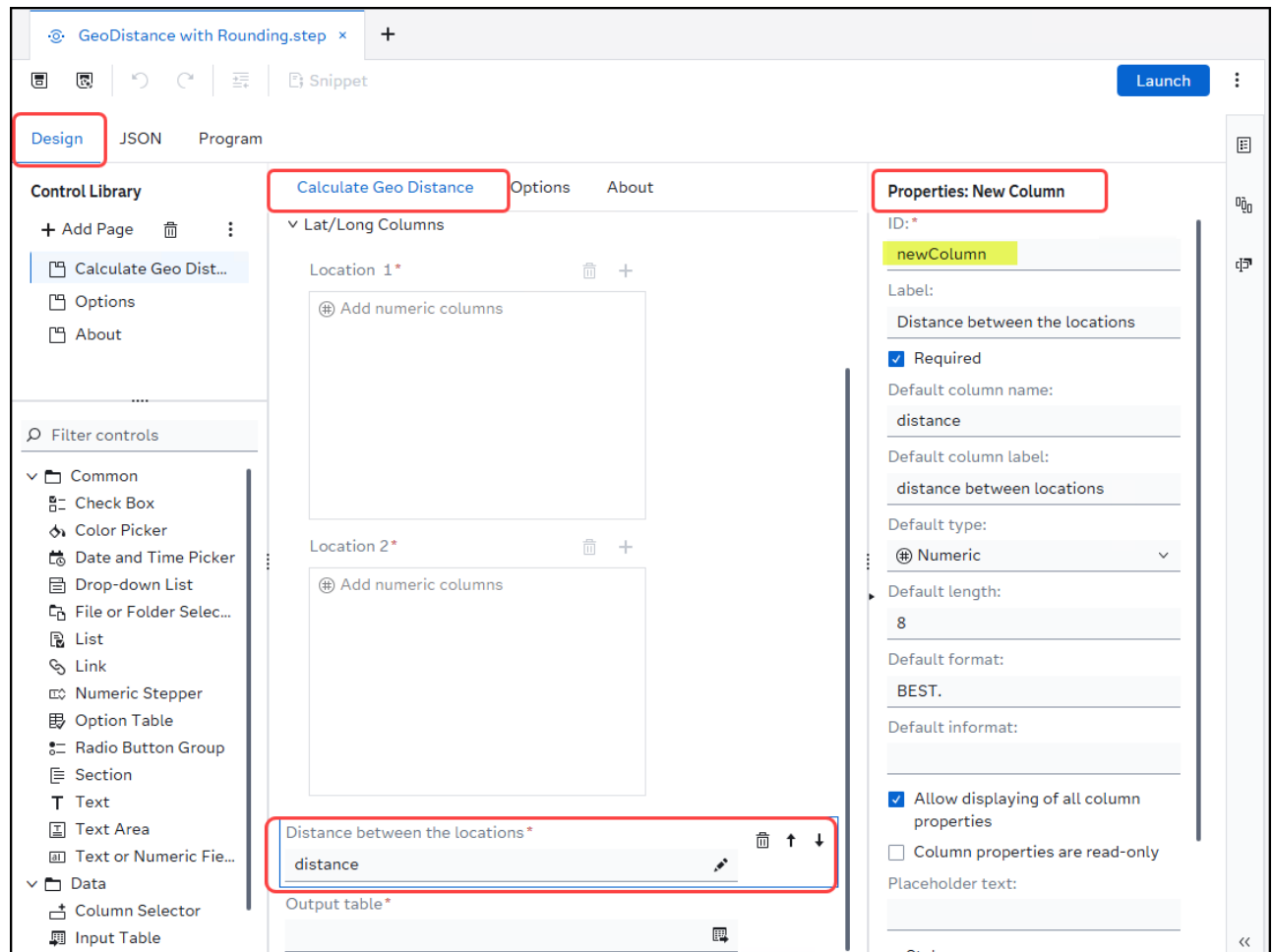
1 %global g_disType;
2 %let g_disType=&disType;
3 %put The value of disType is: &disType;
4
5 /* Set Input and Output Data */
6 data &outTable ;
7     set &inTable ;
8
9 /* newColumn is a hidden control in the UI. The code sets the name of the new output column. */
10
11 /* Calculate GeoDistance in Miles */
12 %if &disType=Miles %then %do ;
13     %let newColumn_name=distance_in_Miles ;
14     &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'M') ;
15 %end;
16
17 /* Calculate GeoDistance in Kilometers */
18 %else %do ;
19     %let newColumn_name=distance_in_Kilometers ;
20     &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'K') ;
21 %end ;
22
23
24 /* Round GeoDistance Value */
25 %if &round=1 %then %do ;
26     select(&decimalPlaces) ;
27         when(0) &newColumn_name=round(distance_in_&disType) ;
28         when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
29         when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
30         when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
31         when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
32         when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
33     end;
34 %end;
35 run;

```

Now, the global macro variable **&g_disType** can be referenced outside of the custom step in a SAS Studio flow.

8. Use the New Column Control (if your code creates a new column)

1. Select the **Design** tab to view the design design(s) for the custom step.
2. Select the **Calculate Geo Distance** page tab.
3. Select the **Distance between the locations** *new column* control to view its properties.

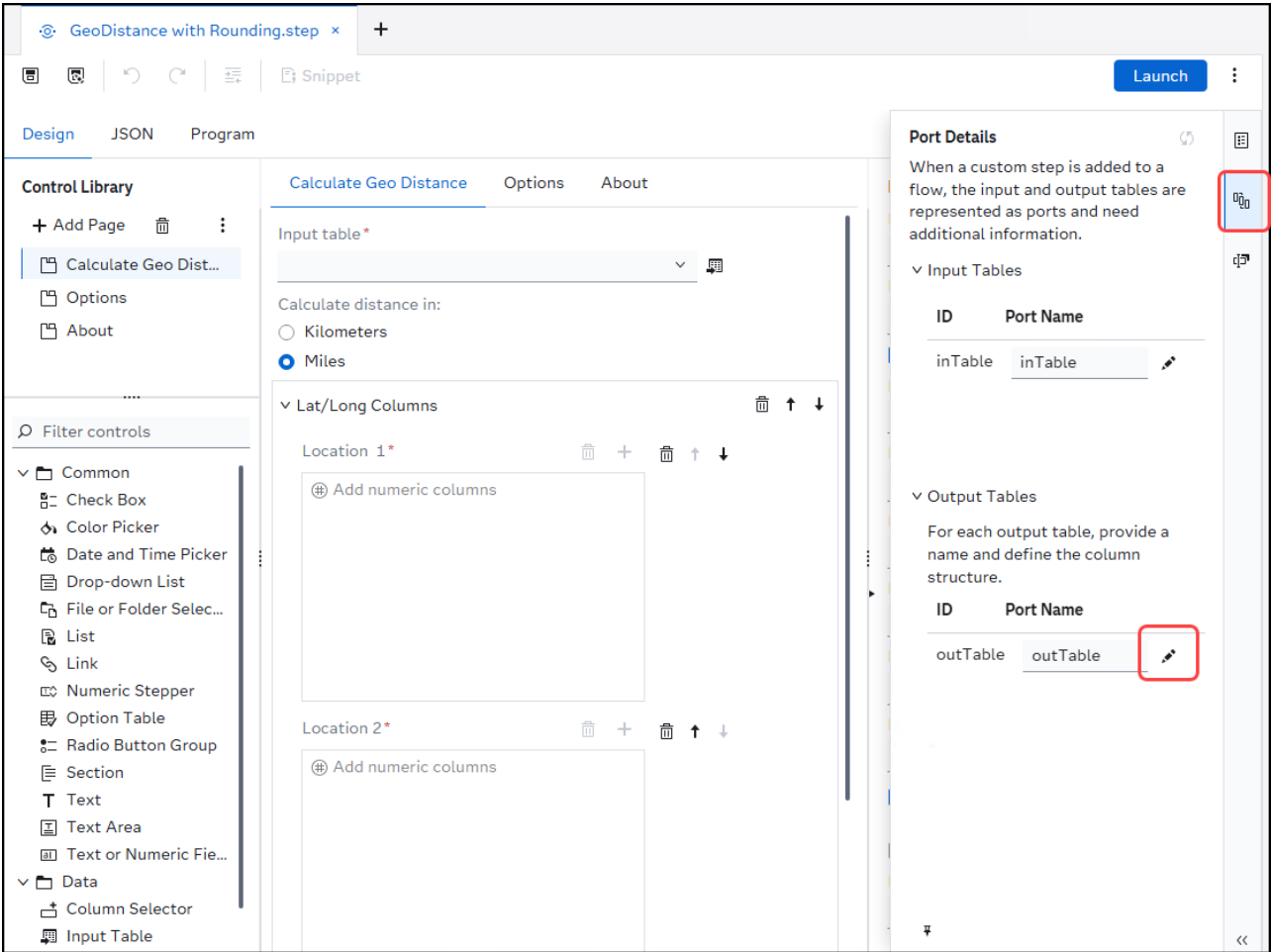


✎ The ID name for the control is **newColumn**.

9. Use Port Details to Control the Output Columns

1. Click  to view the port details for the custom step.

2. Select  in the *Output Tables* section to view the output table structure for the custom step.



3. All the columns from the input table are part of the output in addition to the new column *Distance between the locations*.

Output Table Metadata

Details

ID:
outTable

Port name: *
outTable

Description:

Define the column structure of the output table

Drop columns ▾ Keep columns ▾ New column ▾ All columns ▾

Data Controls	Linked Input Control	Publish
Input table ID: inTable		All
Distance between the locations ID: newColumn		New

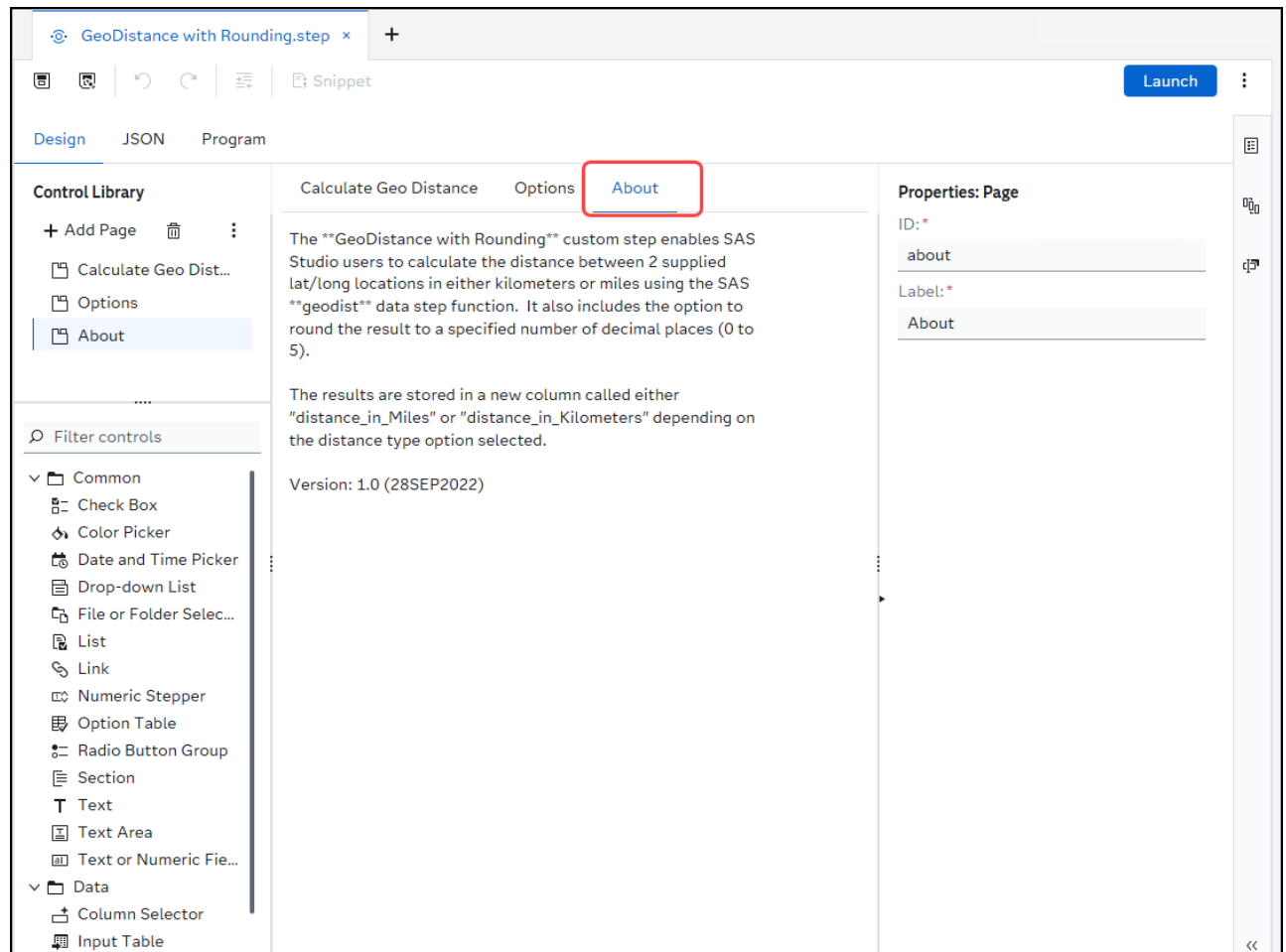
OK Cancel

✎ If you decide to drop columns from the output table, then you should also explicitly drop the columns in the Program code as well.

4. Click **Cancel** to close the *Output Table Metadata* dialog without making any changes.

10. Create an **About** tab for the Custom Step

1. Click the **About** page tab for the custom step to view its properties. It is a best practice to include a *Text* control on this page with information about what the step does and its version information on this tab.



To create a new page for an *About* tab, select **+ Add Page** in the **Control Library** section on the *Design* tab for the custom step.

2. Click **x** to close the *GeoDistance with Rounding.step* tab.

Exercise Completed

You have completed the exercise for **Tips & Tricks for Building a Custom Step with SAS Viya**.

THANK YOU FOR ATTENDING THIS SESSION!

PLEASE COMPLETE THE EVALUATION!

Custom Steps

Resources



Training Course

- [Developing Custom Steps with SAS® Studio Analyst](#)

Documentation

- [SAS Help Center: SAS Studio: Working with Custom Steps](#)

SAS Community Articles

- [Search - SAS Support Communities for Custom Steps](#)

Quick Start Video

- [Create Custom Steps with SAS Studio](#)

YouTube Video Series

- [Creating a Custom Step in SAS Studio](#)

GitHub Repository of Custom Steps

- <https://github.com/sassoftware/sas-studio-custom-steps>

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Training Course:

- [Developing Custom Steps with SAS® Studio Analyst:](#)
<https://learn.sas.com/course/view.php?id=220>

Documentation:

- [SAS Help Center: SAS Studio: Working with Custom Steps:](#)
<https://go.documentation.sas.com/doc/en/sasstudiocdc/default/webeditorcdc/webeditorsteps/titlepage.htm>

SAS Community Articles:

- [Search - SAS Support Communities for Custom Steps:](#)
https://communities.sas.com/t5/forums/searchpage/tab/message?q=Custom%20Steps&noSynonym=false&collapse_discussion=true

Quick Start Video:

- [Create Custom Steps with SAS Studio:](#)
<https://video.sas.com/detail/video/6347011003112/create-custom-steps-with-sas-studio>

YouTube Video Series:

- [Creating a Custom Step in SAS Studio:](#)
<https://youtube.com/playlist?list=PLVV6eZFA22QyLhfnKKW3k08XVx1KOZCJ4&si=htu2O3BmS05oo9aD>

GitHub Repository of Custom Steps:

- <https://github.com/sassoftware/sas-studio-custom-steps>

Thanks for attending this Hands-on Workshop session!

Please complete the evaluation for the session.

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